Supplemental material for

Deep learning for denoising 3D sonar point clouds in underwater scour investigation and infrastructure inspection

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1 Extended validation experiments for the equivalent thickness metric

This section describes additional validation experiments for the proposed equivalent thickness metric, with a focus on validating its effectiveness for infrastructure-inspection targets, analyzing the physical relationship between the equivalent thickness metric and P2S, and clarifying the physical meaning of the nominal noise intensity.

1.1 Equivalent-thickness validation instantiated for infrastructure-inspection targets

In this extended validation study, the simulated noise conditions and evaluation metrics were kept consistent with those used in the main-text validation experiments. 400 cylindrical point clouds with heights and radii randomly sampled from 5 to 25 were generated to emulate infrastructure-inspection targets. Representative results were shown in Fig. 1 (a). The results of the extended experiments were consistent with those of the main-text validation in underwater terrain scenarios. Equivalent thickness was strongly correlated with the simulated noise level, widely used computational metrics (CD and HD), and downstream task metrics (volumetric error and projected area error), as shown in Fig. 1 (b). Therefore, it could serve as a surrogate metric in real-world engineering settings where ground truth is unavailable, replacing ground-truth-dependent measures such as CD and HD to assess noise-induced degradation in point cloud quality.

1.2 Physical relationship between equivalent thickness and P2S

To further examine the relationship between the proposed metric and P2S, the simulated Gaussian noise levels were refined, and 100 additional piles were generated with heights and radii uniformly sampled between 5 and 25. The above validation procedure was then repeated. A scatter plot

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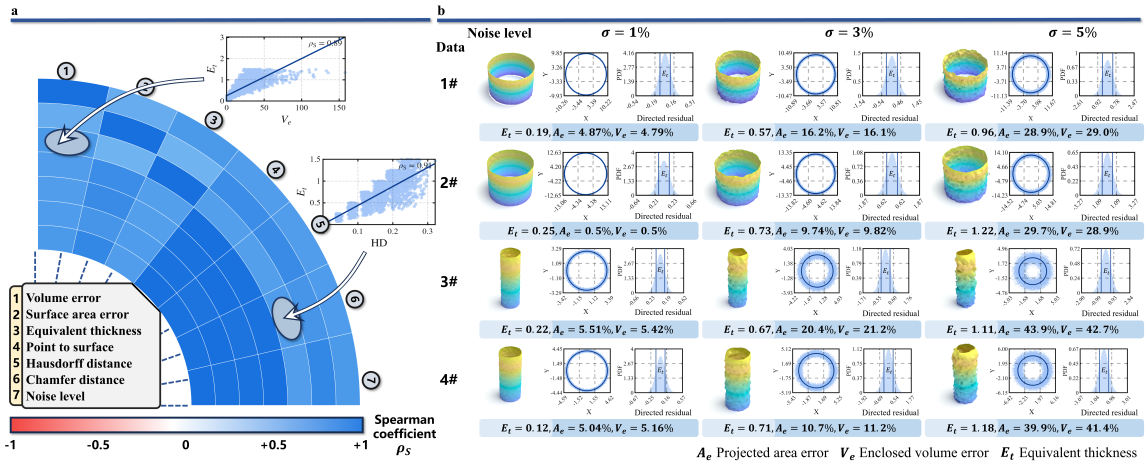

Figure 1: Validation of the proposed equivalent thickness metric. a Correlation analysis among all experimental metrics. b Visualisation of representative results from the experiments.

of the mean absolute directional residual radius versus P2S was shown in Fig. 2. Across all experiments, the mean difference between P2S and the residual radius was 1.6394×10^{-5}, and the fitted slope was 0.997, indicating that the two measures were numerically equivalent. This result supported the statement above: when $\mathbf{u}$ in Eq.(21) of the main text was defined as the normal vector of the manifold-like geometric approximation $\mathcal{M}$ of the simulated pile sonar point cloud, the directional residual was physically consistent with P2S.

In addition, the relationship between the noise level and the proposed metric was examined by analyzing the dependence of P2S on equivalent thickness. A scatter plot of P2S versus one-half of the equivalent thickness values exhibited an approximately linear trend, with a slope close to 1.252, i.e., the reciprocal of 0.797. Based on the above analysis, P2S and the equivalent thickness were interpreted as two statistical descriptions of the same underlying variable, i.e., the directional residual: specifically, the mean absolute value and the standard deviation, respectively. When Gaussian noise was added, the following relationship was obtained:

$$r(\mathbf{f}) \approx \mathbf{u}^{\mathsf{T}}(\mathbf{e}_f + \mathbf{e}_u) \sim \mathcal{N}\left(0, \sigma^2\right) \tag{1}$$

$$\mathrm{P2S} = \mathbb{E}(|r|) \tag{2}$$

$$E_t = 2\sqrt{\mathbb{E}[(r - \mathbb{E}(r))^2]} \tag{3}$$

$$\mathrm{P2S} = \sqrt{\frac{2}{\pi}} \times \sqrt{\mathbb{E}[(r - \mathbb{E}(r))^2]} = 0.797884\frac{E_t}{2} \tag{4}$$

The above derivation was consistent with the experimentally observed slope, further corroborating the physical interpretability of the proposed equivalent thickness metric. In practice, the equivalent thickness was used to quantify the extent to which a given type of noise degraded the structural surface of a point cloud, and it was regarded as a statistical manifestation on a lower-dimensional manifold of noise-induced corruption of the high-dimensional ground truth geometry. Because accurately estimating the three-dimensional noise-induced morphological distortion from real measured data was highly challenging, an equivalent-thickness-based strategy was adopted. Specifically, by leveraging prior geometric characteristics of the target, a referenceable approximate estimate was constructed on the low-dimensional manifold, thereby providing a feasible pathway for analyzing complex three-dimensional noise patterns and their intensities.

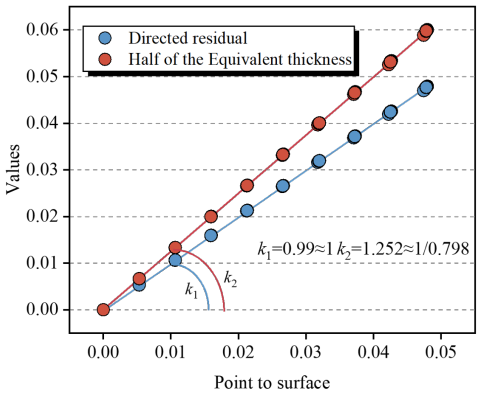

Figure 2: Relationship between equivalent thickness, directional residuals, and P2S.

1.3 Nominal noise level and its physical interpretation

To enable a more objective assessment of the noise level in measured point clouds, a derived indicator was proposed based on the equivalent thickness. From the field observations, the local surfaces of structures with different characteristic sizes were degraded to different extents under the same noise intensity. Therefore, to eliminate the influence of the length unit (meters) of E_t on the analysis, a nominal noise level, on the analysis, a nominal noise level, ε_n, was defined using the equivalent thickness and the diameter of the point cloud's normalized bounding sphere, denoted as L:

$$\varepsilon_n = \frac{E_t}{L} \tag{5}$$

In the above experiments, a scatter plot was also generated between the nominal noise level ε_n and the actual noise level specified in the experimental settings as shown in Fig. 3. It was observed that, at low noise levels, ε_n closely matched the simulated noise intensity. As the noise increased, however, ε_n tended to underestimate the ground-truth value. This bias was attributed to the fact that, relative to the clean point cloud, added noise enlarged the absolute geometric extent (contour size) of the point cloud, thereby increasing the diameter of the bounding sphere used for normalization. Consequently, the normalized point cloud was effectively compressed, leading to a nominal noise level smaller than the true value, with the effect becoming more pronounced at higher noise levels. For example, at a noise level of 6%, the ratio was approximately 0.8, while ε_n still provided a reliable estimate of the noise intensity.

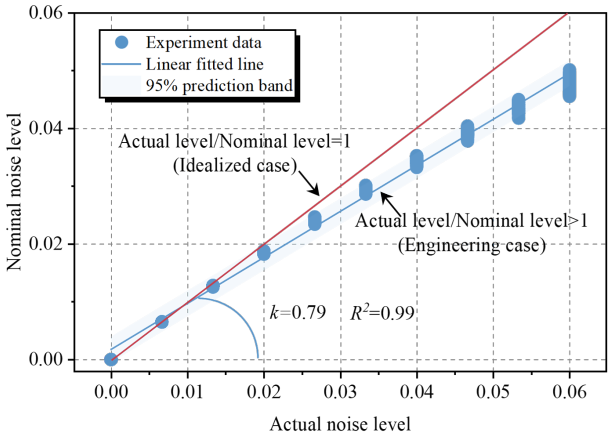

Figure 3: Relationship between the nominal noise level and the actual noise level.

1.4 Robustness experiments and analysis

In this section, the robustness of the proposed equivalent-thickness metric is systematically evaluated through three sets of experiments. First, noise generalization experiments are conducted under non-Gaussian noise and other noise patterns containing large, sparsely distributed outliers. Second, morphological robustness is evaluated using point clouds containing structural damage or local defects, and the influence of surface curvature on the metric is further investigated. Third, parameter sensitivity analysis is performed to assess the effects of interpolation density and window size on the computation of the equivalent-thickness metric.

1.4.1 Noise generalization analysis

For the noise generalization experiments, the half-ellipsoidal point clouds and parameter ranges described in Section 5.1 of the main text were used to simulate underwater terrain, whereas the cylindrical point clouds and parameter ranges described in Section 1 of this Supplemental Material were used to simulate underwater bridge piles. One hundred point clouds were generated for each geometric category.

Four noise distributions were considered: t-distributed noise at contamination levels of 1–6% with 5–10 degrees of freedom (DoF), uniform noise at levels of 1–6%, discrete noise at levels of 1–6%, and Laplacian noise at levels of 1–6%. t-distributed noise and discrete noise were used to reproduce the heavy-tailed noise patterns and large, sparsely distributed outliers observed in the preceding noise-morphology analysis, respectively. Common geometric metrics, including Chamfer distance (CD), Hausdorff distance (HD), and point-to-surface distance (P2S), were calculated. The area error A_e and volume error V_e, which quantified the interference of noise with downstream tasks, were also evaluated, together with the noise intensity σ and equivalent thickness E_t. Spearman's rank correlation coefficients were then computed to quantify the associations among these variables. The noise generalization results of the equivalent-thickness metric for underwater scour terrain and bridge foundation inspection are presented in Fig. 4.

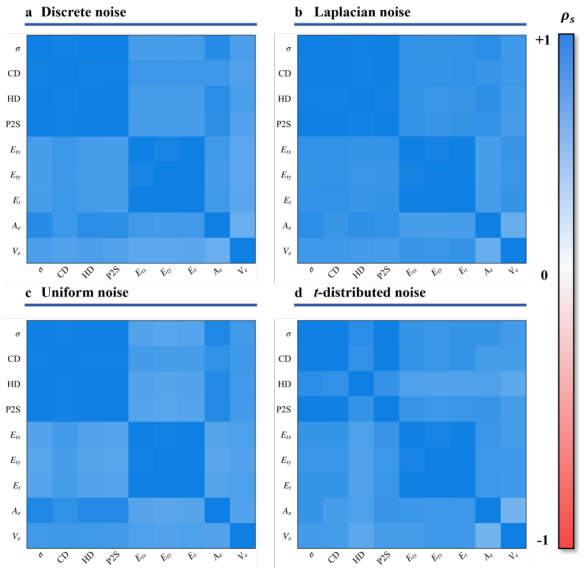

Figure 4: The noise generalization results of the equivalent thickness metric for underwater scour terrain inspection a. The noise generalization results under Discrete noise. b. The noise generalization results under Laplacian noise. c. The noise generalization results under Uniform noise. d. The noise generalization results under t-distributed noise.

Fig. 4 and 5 present the Spearman rank correlation matrices for the scour-terrain and bridge-foundation scenarios under four noise patterns (uniform, Laplacian, discrete, and heavy-tailed). Across all cases, the equivalent-thickness metric E_t exhibited consistently strong positive

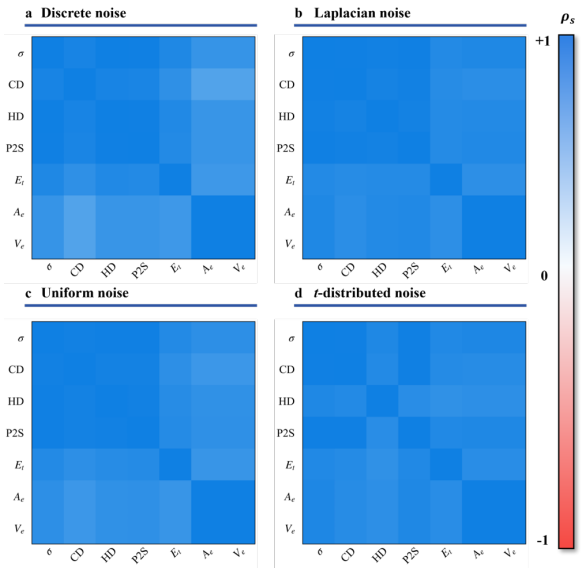

Figure 5: The noise generalization results of the equivalent thickness metric for underwater bridge foundation inspection. a. The noise generalization results under Discrete noise. b. The noise generalization results under Laplacian noise. c. The noise generalization results under Uniform noise. d. The noise generalization results under t-distributed noise.

associations with the geometric error measures and the noise intensity. For the bridge-foundation point clouds, the Spearman coefficients between E_t and σ, CD, HD, and P2S remained high under all noise types (typically $\rho_S \approx 0.86 - 0.93$). For the scour-terrain point clouds, E_t (represented by the directional thickness components and their mean) likewise showed stable positive correlations with σ, CD, HD, and P2S ($\rho_S \approx 0.70 - 0.86$), although the magnitudes were moderately lower than those observed for the foundation cases. These results indicate that E_t retains a clear physical interpretation under non-Gaussian and outlier-prone noise: larger equivalent thickness corresponds to stronger geometric deviation from the clean surface, in agreement with conventional distance-based metrics.

Meanwhile, E_t was also strongly correlated with the area and volume errors that characterize noise-induced degradation in downstream geometric tasks. In the foundation experiments, the associations between E_t and A_e/V_e reached $\rho_S \approx 0.80 - 0.89$; in the terrain experiments, the corresponding coefficients remained substantial ($\rho_S \approx 0.69 - 0.83$). Importantly, these high correlations persisted across all tested noise morphologies, demonstrating that E_t not only aligns with classical geometric discrepancy measures but also reliably reflects the adverse impact of noise on area- and volume-related downstream quantities. Overall, the noise-generalization analysis confirms that the proposed equivalent-thickness metric is robust to diverse noise distributions and provides a physically meaningful, task-relevant indicator of point-cloud quality.

1.4.2 Morphological robustness analysis

This subsection evaluates the morphological robustness of the proposed equivalent-thickness metric from two complementary perspectives. First, its robustness in the presence of local defects or structural damage is assessed by examining how the correlations between equivalent thickness and other evaluation metrics vary with the severity of the simulated damage. Second, the relationship between surface curvature and equivalent thickness is investigated to evaluate the generalizability of the metric across inspection targets with different curvatures.

Specifically, to investigate the influence of actual structural damage on point-cloud-based assessment, localized surface defects were introduced into the simulated bridge-foundation point clouds. The simulation parameters and data-generation procedure were kept consistent with

those described in Section 5.1 of the main text, and a total of 100 point cloud realizations were generated. Five defect severity levels, denoted as Levels I-V, were considered. These levels corresponded to normalized defect depth ratios $\eta_d = d_{\max}/R$, of 0, 0.12, 0.24, 0.36, and 0.48, respectively, where $d_{\max}$ denotes the maximum local defect depth and R is the foundation radius. Level I represented the undamaged control condition, while Levels II-V represented progressively increasing degrees of local damage. In particular, Level V corresponded to severe localized deterioration and was intended to approximate extreme defect conditions that may be encountered during field inspections. The simulated bridge-foundation point clouds containing structural defects are shown in Fig. 6.

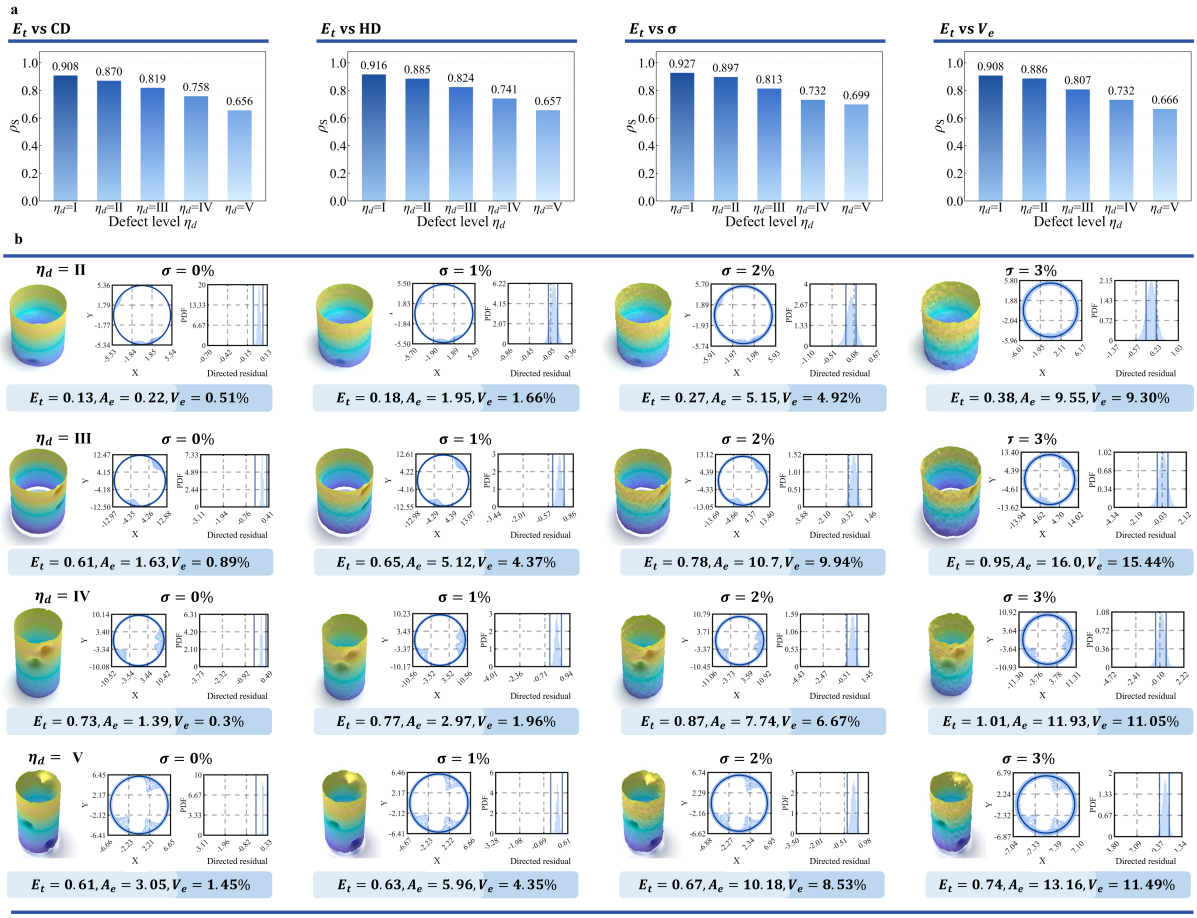

Figure 6: Morphological robustness analysis considering point clouds with local defects or structural damage. a. Comparison of correlations across different defect severity levels. b. Visualizations of point clouds containing defects or structural damage at different severity levels.

For each defect level, the equivalent thickness E_t was computed together with Chamfer distance (CD), Hausdorff distance (HD), area error A_e, volume error V_e, and noise intensity σ. Spearman's rank correlations between E_t and these quantities were then evaluated after pooling samples across noise levels within each η_d. As shown in Fig. 6, E_t remained strongly and positively correlated with σ, CD, HD, A_e, and V_e under all defect levels. For the undamaged control ($\eta_d = \mathrm{I}$), the Spearman coefficients were typically in the range of approximately 0.91–0.93. Under defect levels representative of commonly encountered engineering scales ($\eta_d = \mathrm{I} - \mathrm{III}$), the Spearman correlations between E_t and these quantities remained above 0.8. As the defect severity increased, these associations decreased gradually; nevertheless, even under the most extreme condition considered here ($\eta_d = \mathrm{V}$), corresponding to a defect depth ratio of 0.48, the coefficients remained above 0.65 and approached 0.70.

These results indicate that local structural damage does not substantially undermine the noise-related interpretability of equivalent thickness. Even in the presence of geometric damage that could interfere with fitting, E_t continued to track the noise intensity and the resulting geometric and downstream-task errors, rather than being confounded by the defects.

A further curvature-based morphological analysis was conducted to evaluate the generalizability of the equivalent-thickness metric across inspection targets with different surface-curvature profiles. Following the same geometric generation strategy as in Section 1 of this Supplemental Material, the 125 hemi-ellipsoidal point clouds were each perturbed with Gaussian noise at nine levels from 0–6% in steps of 0.75% to generate noisy point clouds. For each realization, mean absolute curvatures were computed along the X- and Y-direction profile centerlines, and the corresponding directional equivalent thicknesses E_{tx} and E_{ty} were obtained. The relationships between mean absolute curvature and equivalent thickness are presented in Fig. 7.

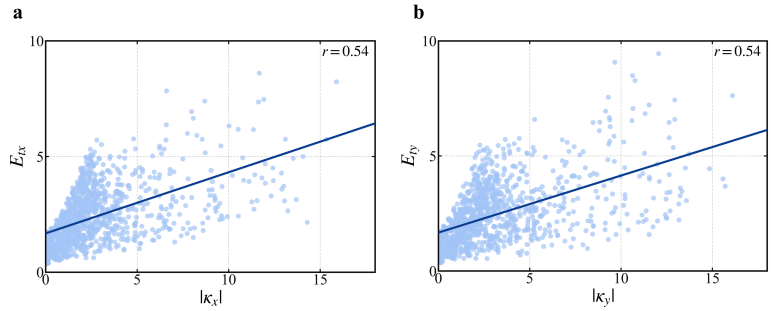

Figure 7: Curvature-based morphological robustness analysis. a Analysis results for the central profile along the X-direction. b Analysis results for the central profile along the Y-direction.

Pearson correlation analysis yielded only moderate associations of $r = 0.543$ (X direction) and $r = 0.544$ (Y direction). These values are substantially lower than the strong correlations previously observed between E_t and noise-related geometric indicators (typically above 0.85 in the foundation cases). These observations suggest that local curvature is not a dominant confounding factor for the equivalent-thickness metric: variations in E_t are primarily governed by noise-induced profile thickening rather than by the intrinsic curved geometry of the underlying surface.

Overall, the morphological robustness analysis demonstrates that the proposed metric remains stable under severe local defects and exhibits limited sensitivity to surface curvature. These results demonstrated that the proposed equivalent-thickness metric exhibited strong robustness for noise assessment across different engineering inspection targets.

1.4.3 Parameter sensitivity analysis

This subsection examines the sensitivity of the equivalent-thickness computation to two implementation parameters: the interpolation density along each profile and the local window size used to evaluate thickness. Both analyses were conducted on the same class of simulated hemi-ellipsoidal point clouds as in Section 5.1 of the main text, with 100 realizations. For each parameter setting, the associations between the equivalent thickness E_t and Chamfer distance (CD), Hausdorff distance (HD), noise intensity σ, area error A_e, and volume error V_e were quantified using Spearman's rank correlation coefficients.

Specifically, window fractions ranging from 1.6% to 12.8% of the profile length were tested as shown in Fig. 8. Overall, smaller windows yielded stronger Spearman correlations between E_t and the reference indicators. At the smallest window fraction (1.6%), the correlations with CD, HD, and σ generally exceeded 0.90, and those with A_e and V_e also remained high. When the window fraction was increased to approximately 3%, the correlations were still strong (typically $\rho_S \approx 0.76 - 0.87$) and retained a clear physical interpretation. Beyond this level,

however, the associations declined steadily.

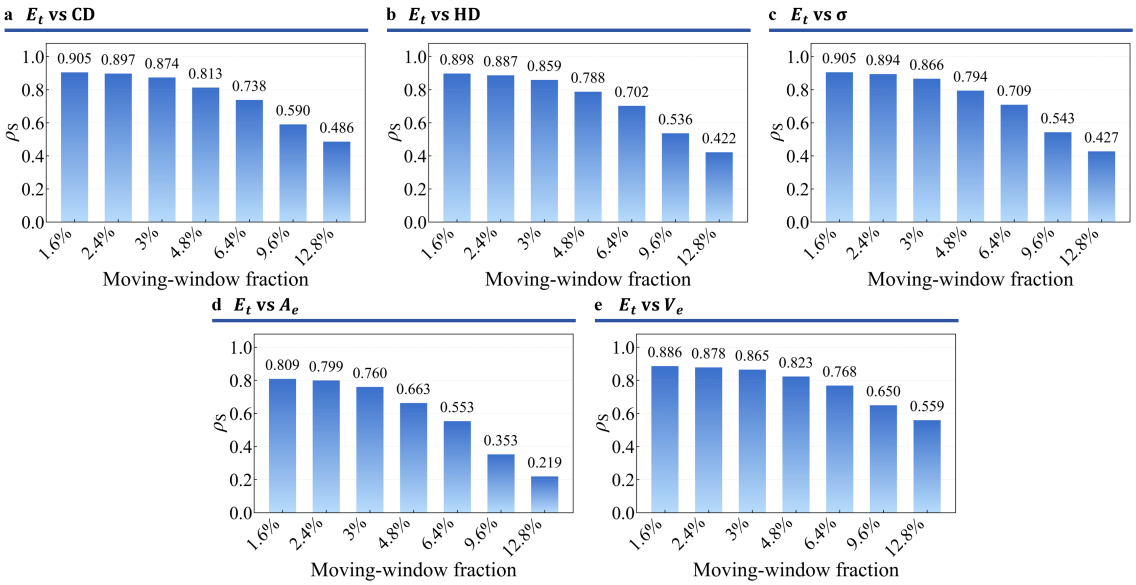

Figure 8: Sensitivity analysis of the equivalent thickness metric to the local window-size parameter. a. Correlation between equivalent thickness and Chamfer distance. b. Correlation between equivalent thickness and Hausdorff distance. c. Correlation between equivalent thickness and noise intensity. d. Correlation between equivalent thickness and area error. e. Correlation between equivalent thickness and volume error.

These results indicate that an excessively large window weakens the noise-related meaning of equivalent thickness. A plausible explanation is that a larger window incorporates a broader portion of the profile trend and is therefore more susceptible to interference from the overall surface curvature, which dilutes the local noise-induced thickening signal. Considering both correlation strength and computational cost, a window size of about 3% of the profile length is recommended for practical implementation.

Furthermore, interpolation densities of 100 to 1300 points per profile were further examined as shown in Fig. 9. In contrast to the window-size trend, increasing the number of interpolation points systematically strengthened the Spearman correlations between E_t and CD, HD, σ, A_e, and V_e. At 100 points, the associations were only weak to moderate (e.g., $\rho_S \approx 0.14$–0.51). As the density increased to 300 and then 500 points, the coefficients rose markedly and approached a high plateau (approximately 0.75–0.86 at 500 points). Further densification beyond 500 points produced only marginal improvements.

This behavior suggests that insufficient interpolation may under-resolve local geometric features, thereby weakening the physical content of the estimated equivalent thickness. Once the profile is sampled at a sufficiently fine resolution, additional points bring diminishing returns. Accordingly, an interpolation density of 500 points per profile is recommended as a practical trade-off between accuracy and computational efficiency.

Overall, the parameter sensitivity analysis shows that, although the equivalent-thickness metric is influenced by both window size and interpolation density, its physical interpretation and performance remain stable across a broad parameter range, specifically when the window size is below 3% of the profile length and the interpolation density exceeds approximately 500 points. Setting the hyperparameters within this range therefore enables robust noise assessment in field applications.

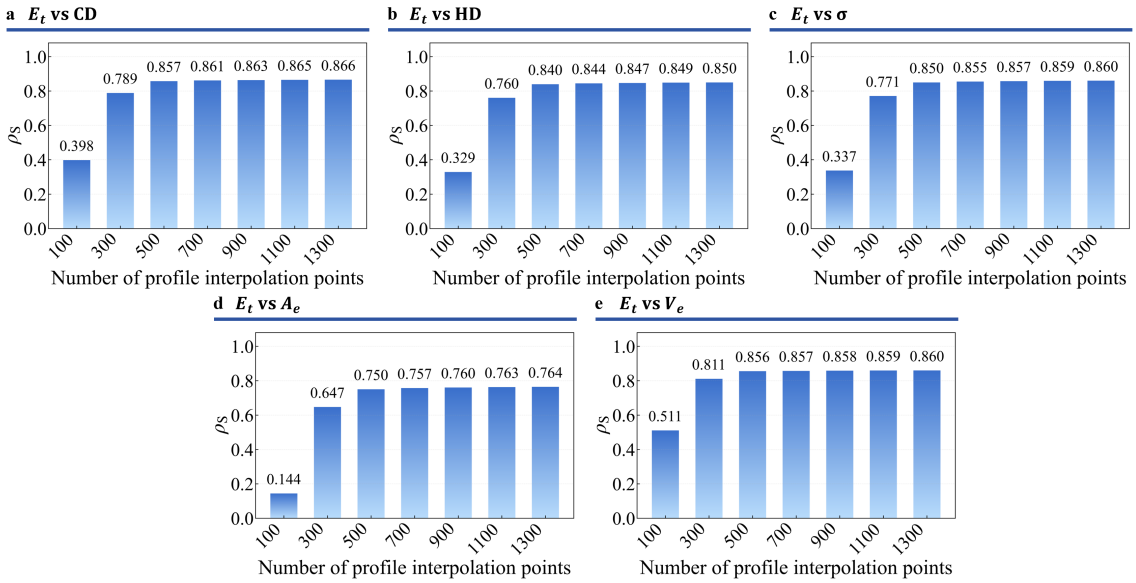

Figure 9: Sensitivity analysis of the equivalent thickness metric to the interpolation density parameter. a. Correlation between equivalent thickness and Chamfer distance. b. Correlation between equivalent thickness and Hausdorff distance. c. Correlation between equivalent thickness and noise intensity. d. Correlation between equivalent thickness and area error. e. Correlation between equivalent thickness and volume error.

2 Implementation details and hyperparameter settings of baselines

This subsection described the hyperparameter settings and tuning procedures used for the baseline methods. All learning-based baseline methods were retrained on the USPCDenoise training set using the source code released by their original authors. The same training, validation, and test splits, input normalization procedure, point-cloud size, and noise-generation strategy were adopted for all methods. The principal hyperparameters of each learning-based baseline were set according to the recommendations of the original authors, except for the batch size and learning rate, which were adjusted when necessary to accommodate GPU memory constraints and ensure training stability. To ensure a fair comparison, the same data-augmentation strategy used for SIREND-Net was applied to all learning-based baselines. Specifically, data augmentation was performed using random rotations of up to 20° about an arbitrary axis and isotropic scaling with a randomly sampled scale factor in the range of [0.95, 1.05].

For the non-learning-based baselines requiring parameter tuning, Gaussian noise at levels ranging from 1% to 6% was added to the USPCDenoise validation set, and an iterative grid-search procedure was performed using the Chamfer distance as the optimization criterion. For Guide, the candidate values of radius were 0.001, 0.01, 0.1, and 1, whereas those of epsilon were 0.08, 0.10, 0.12, and 0.14. For WLOP, gridStep was searched over 0.003, 0.03, 0.3, and 3, and the repulsion coefficient was searched over 0.30, 0.35, 0.40, and 0.45. For SOR, nb_neighbors was searched over 16, 18, 20, 22, and 24, whereas std_ratio was searched over 0.1, 0.2, 0.3, 0.4, and 0.5. For ROR, nb_neighbors was searched over 6, 8, 10, 12, and 14, and radius_factor was searched over 1.6, 1.8, 2.0, 2.2, and 2.4. The selected parameter settings and the hyperparameters used for the other baseline methods are summarized in the table below.

Table 1: Detailed hyperparameter settings of the baseline methods.

Method (Types)	**Local training or algorithm settings**
DMR (Learning-based)	1) EdgeConv KNN = $\{8, 16, 24\}$; 2) Mesh_steps: 2; 3) The folding decoder channels use 313–128–128–64–3; 4) downsampling ratio = 0.5; 5) Random seed = 2025; 6) Training epoch: 400 epochs; 7) batch size: 4; 8) Optimizer: Adam; 9) Learning rate = 5×10^{-4}; 10) Scheduler: ReduceLROnPlateau (factor = 0.5, patience = 10; min_lr = 1×10^{-6}).
Score (Learning-based)	1) Batch size = 6; 2) frame_knn = 32; 3) num_train_points = 128; 4) num_clean_nbs = 4; 5) num_selfsup_nbs = 8; 6) dsm_sigma = 0.01; 7) Training epoch: 400 epochs; 8) ScoreNet hidden dimension = 128; four residual blocks; 9) Optimizer: Adam; 10) Learning rate = 1×10^{-4}, weight decay = 0; 11) Random seed = 2025.
MODNet (Learning-based)	1) Batch size = 64; 2) Training epoch: 100 epochs; 3) Optimizer: SGD; 4) Learning rate = 1×10^{-4}, momentum = 0.9; 5) patch_radius = 0.03; 6) support_multiple = 4; 7) support_angle = 15°; 8) repulsion_alpha = 0.97; 9) Random seed = 2025; 10) For epochs > $\{8, 14, 18, 24, 27\}$, learning-rate multipliers are $\{0.1, 0.05, 0.01, 0.005, 10^{-4}\}$.
PCDNF (Learning-based)	1) patch_radius = 0.05; 2) Batch size = 200; 3) Training epoch: 50 epochs; 4) Optimizer: SGD; 5) Learning rate = 1×10^{-4}, momentum = 0.9; 6) support_multiple = 4; 7) support_angle = 15°; 8) repulsion_alpha = 0.98; 9) Random seed = 2025; 10) Internal KNN: $k = 8/16$ for feature extraction; $k = 10$ for feature fusion. 11) For epochs > $\{16, 24, 32, 36\}$, learning-rate multipliers are $\{0.1, 0.01, 0.001, 0.0005\}$.

Table 1: Detailed hyperparameter settings of the baseline methods (continued).

Method (Types)	**Local training or algorithm settings**
PCNet (Learning-based)	1) Training epoch: 500 epochs; 2) Batch size = 64; 3) patch_radius = [0.05]; 4) Random seed = 2025; 5) Optimizer: SGD; 6) Learning rate = 1×10^{-8}, momentum = 0.9; 7) Max symmetric operator and point_tuple = 1.
Pointfilter (Learning-based)	1) patch_radius = 0.05; 2) Batch size = 64; 3) Training epoch: 100 epochs; 4) Random seed = 2025; 5) Optimizer: SGD; 6) Learning rate = 1×10^{-4}, momentum = 0.9; 7) support_multiple = 4; 8) support_angle = 15°; 9) repulsion_alpha = 0.97; 10) For epochs > $\{16, 24, 32, 36\}$, learning-rate multipliers are $\{0.1, 0.01, 0.001, 0.0005\}$.
Guide (Traditional)	1) radius = 0.1; 2) epsilon = 0.12; 3) minimum neighborhood size = 3.
WLOP (Traditional)	1) gridStep = 0.03; 2) iterNum = 25; 3) n = 2048; 4) repulsion coefficient = 0.40.
SOR (Traditional)	1) nb_neighbors = 18; 2) std_ratio = 0.3.
ROR (Traditional)	1) nb_points = 8; 2) radius_factor = 1.8.

3 USPCDenoise dataset construction procedure

This section presents the complete workflow for constructing the USPCDenoise dataset, together with the associated quality-control criteria. Representative quality-control cases encountered during dataset construction are further provided to demonstrate how ambiguous and challenging scenarios were resolved in practice, thereby improving the transparency and reproducibility of the dataset construction process and facilitating readers' understanding.

3.1 Standardized workflow and quality-control protocol

The USPCDenoise was constructed by experienced technicians with bridge engineering backgrounds. The annotation team consisted of trained master students, doctoral students, and professional inspection personnel. To reduce individual bias, the technicians were divided into three groups, each comprising two members, and all groups followed the same standardised workflow when processing the sonar point clouds.

For pile-foundation point clouds, the raw sonar point clouds were first imported into Cloud-

Table 2: Quality-control criteria for USPCDenoise construction.

No.	Description
Q1	Whether true structural boundaries were preserved.
Q2	Whether evident isolated outliers were removed.
Q3	Whether over-smoothing or excessive point deletion was avoided.
Q4	Whether the remaining point density was sufficient to represent the target surface.
Q5	Whether local cross-sectional profiles were consistent with design drawings or the expected geometric trend.

Compare. The built-in Statistical Outlier Removal (SOR) filter was applied for preliminary removal of isolated outlier noise. The technicians then compared each point cloud with the corresponding bridge foundation design drawings to determine whether attachments, temporary construction components, or other non-structural objects existed outside the pile surface. Ambiguous fuzzy noise and non-structural components were manually removed using the Segment tool in CloudCompare, while the main pile boundary and structural contour were preserved.

For riverbed-terrain point clouds, the raw sonar point clouds were also first processed using the SOR filter in CloudCompare to remove isolated outliers. The technicians then referred to prior bathymetric survey maps and measurement coordinates to determine the overall geometric trend of the riverbed. Fuzzy noise points that were inconsistent with the dominant terrain trend were manually removed using the Segment tool, while continuous terrain morphology and local scour-related geometric variations were preserved. After one group completed denoising, the processed result was cross-checked by the other groups. The quality-control criteria (Q1-Q5) included in Table 2.

Only point clouds satisfying all criteria (Q1-Q5) were included in USPCDenoise as clean references. Samples that failed any criterion were subjected to further review using the original point cloud, design drawings, and local cross-sectional views; depending on the review outcome, they were either re-denoised or excluded from the dataset.

Finally, the denoised and cross-checked point clouds were normalized and downsampled to obtain a consistent data format. Point clouds representing infrastructure-inspection targets were uniformly downsampled to 2048 points, whereas point clouds representing scour-mapping terrain targets were uniformly downsampled to 3072 points. These standardized samples formed the final USPCDenoise dataset.

3.2 Representative quality-control examples

This subsection presents representative quality-control examples used during the construction of USPCDenoise. These examples were independent of the USPCDenoise dataset and were specifically developed to guide technical personnel in noise removal and data screening. They are provided to illustrate how ambiguous and challenging cases encountered during dataset construction were identified. Representative quality-control examples are presented in the figure below. Three representative datasets (labelled 1#, 2#, and 3#) are used to illustrate cases that failed to satisfy Quality Criteria Q1–Q5, together with the corresponding desired denoising outcomes.

Specifically, Criterion Q1 addressed irregular point-cloud boundaries introduced during manual segmentation. For example, the Q1-unmet sample of Data 3# exhibited a noticeable gap caused by manual cropping in CloudCompare. Criterion Q2 targeted incomplete noise removal. As shown in the Q2-unmet samples of Data 1#–3#, a considerable number of outlier points remained after denoising. Criterion Q3 addressed over-filtering, which resulted in over-smoothed point clouds or the unintended removal of valid data. The Q3-unmet sample of Data 2# demonstrated this issue, where repeated application of the built-in CloudCompare

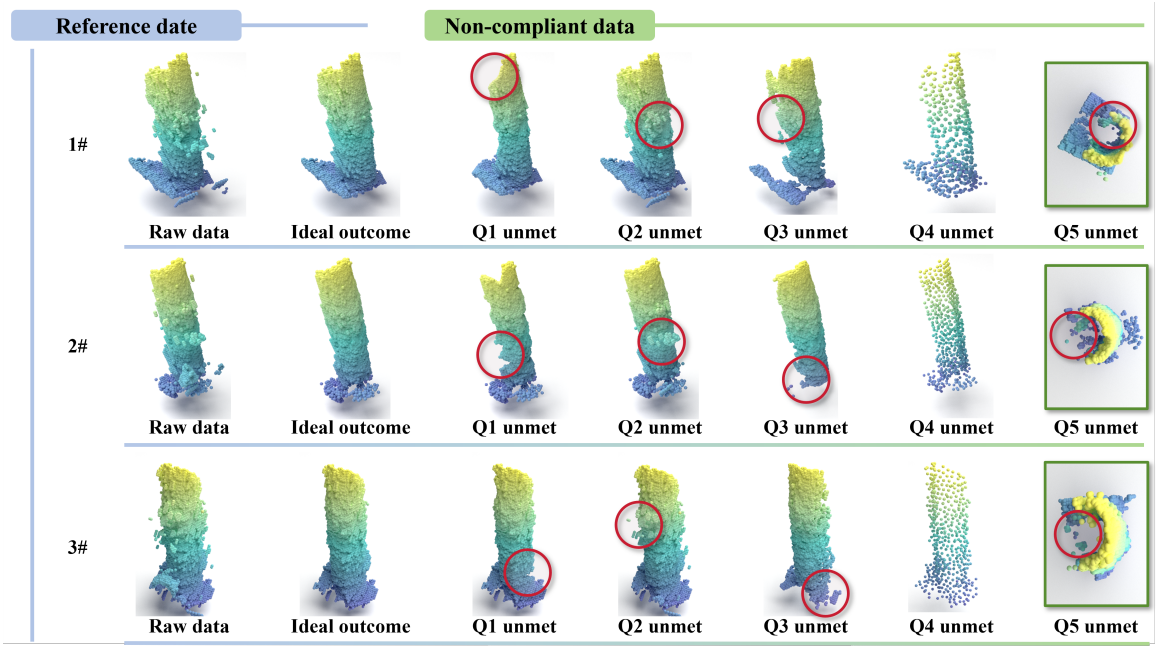

Figure 10: Representative quality-control examples that failed to satisfy each quality criterion

denoising filter incorrectly removed clean points. Criterion Q4 focused on excessive downsampling. As illustrated by the Q4-unmet samples, only approximately 400–600 points were retained, resulting in an insufficient point density for subsequent processing. Criterion Q5 addressed the erroneous removal of valid structures during fuzzy-noise suppression in CloudCompare, leading to substantial deviations from the engineering design. In the Q5-unmet samples of Data 1#–3#, the projected cross-sections perpendicular to the pile axis deviated markedly from the expected circular geometry.

4 Results of the extended numerical performance experiments

This section provides supplementary performance experiments and analyses, including denoising performance analyses based on horizontal projections, P2S changes, and comparisons of two-dimensional boundaries of denoised point clouds, and an evaluation of the effects of the commonly used industrial filtering methods SOR and ROR on point counts. A detailed experiment on the supervised–unsupervised mode switching mechanism.

4.1 Extended qualitative results

Based on representative samples under typical noise conditions shown in Fig. 9 of the main text, the horizontal projections before and after SIREND-Net denoising were further visualized, along with the per-point changes in P2S values. As shown in Fig. 11 (a), noise interference in underwater mapping targets appeared in the 2D projections as speckles on otherwise smooth surfaces. Such artifacts hindered subsequent interpolation, producing unsmooth profiles or surfaces and obstructing downstream mapping applications. After denoising with SIREND-Net, these protrusions and visible irregularities were effectively removed, and the manifold reconstruction restored smooth projections while accurately predicting cross-sections. Moreover, the P2S values computed before and after denoising were also reported in Fig. 11 (b). According to the relationships among P2S, equivalent thickness, and the simulated noise level analyzed in Section 1 of this Supplemental Material, 1.25×P2S was expected to be theoretically close to the simulated noise level. The computed values were largely within 0.031–0.032, which corresponded

to approximately 0.04 after scaling, in close agreement with the 4% noise level used in the simulation.

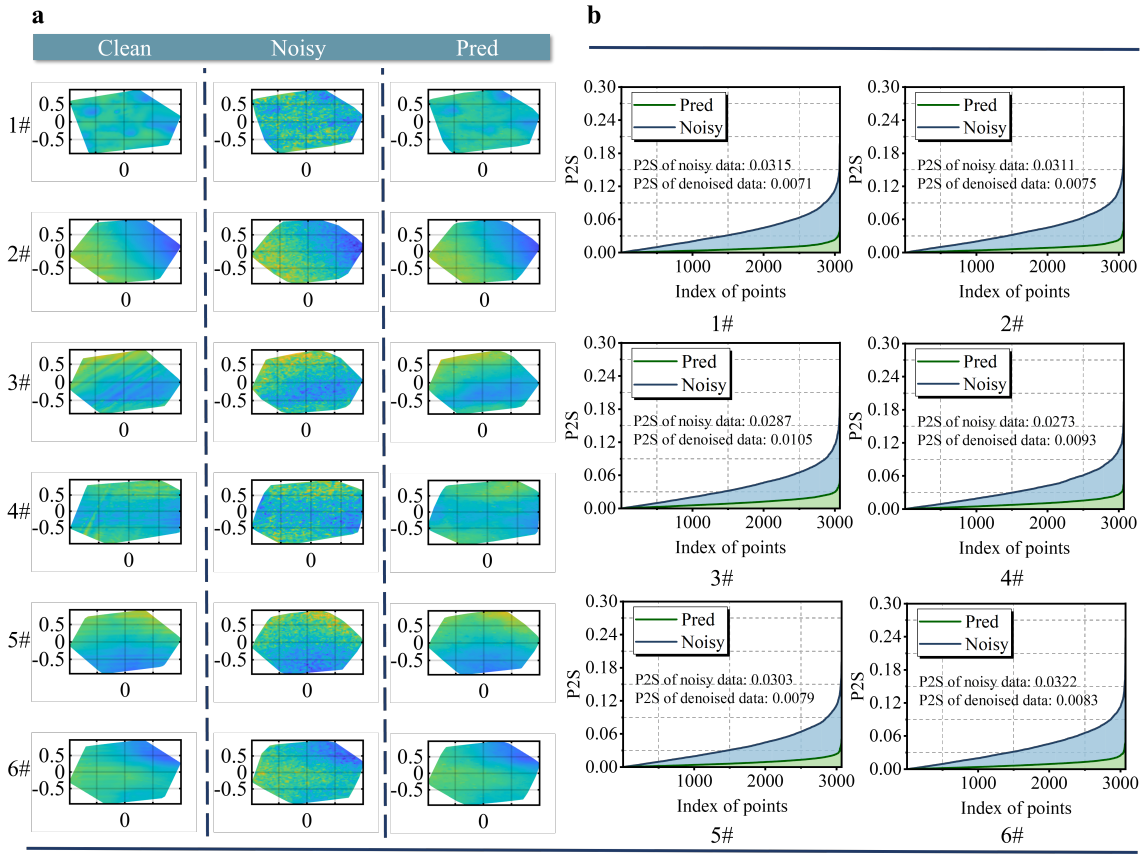

Figure 11: Extended qualitative results of numerical experiments under typical noise condition (4% Gaussian noise). a Comparison of horizontal projections before and after denoising. b Changes in P2S values of individual points before and after denoising, sorted in ascending order.

4.2 Extended quantitative results

In this section, an extended visualization and quantitative analysis were performed for Fig. 12 and Table 3, which compared the extracted 2D boundaries after baseline denoising under typical noise conditions. Further, two metrics were calculated to quantitatively analyze the differences among the baselines in terms of boundary morphology after denoising: (1) Area ratio, defined as the ratio between the enclosed area of the extracted 2D boundary from the denoised point cloud and that from the clean point cloud over the entire test set; and (2) Chamfer distance, computed between the extracted 2D boundarie points of the denoised and clean point clouds. The results were presented in Fig. 13. As shown in the figure, the proposed SIREND-Net achieved the best performance in the supervised setting in preserving geometric details of the denoised point cloud boundaries. The enclosed area of the denoised point clouds was closest to that of the clean point clouds, and the boundary geometry was also the most similar. Moreover, the unsupervised variant also exhibited a clear and competitive advantage. The proposed unsupervised method effectively preserved the principal boundary features and overall contours of the point clouds. This performance was attributed to the tendency of the N2P attention mechanism to focus on boundary points under the unsupervised setting, thereby reducing the risk of erroneous convergence in the noise suppressor.

Consistent with the qualitative observations, the geometry produced by Score deviated markedly, as evidenced by a 30.9% average reduction in the enclosed boundary area of the

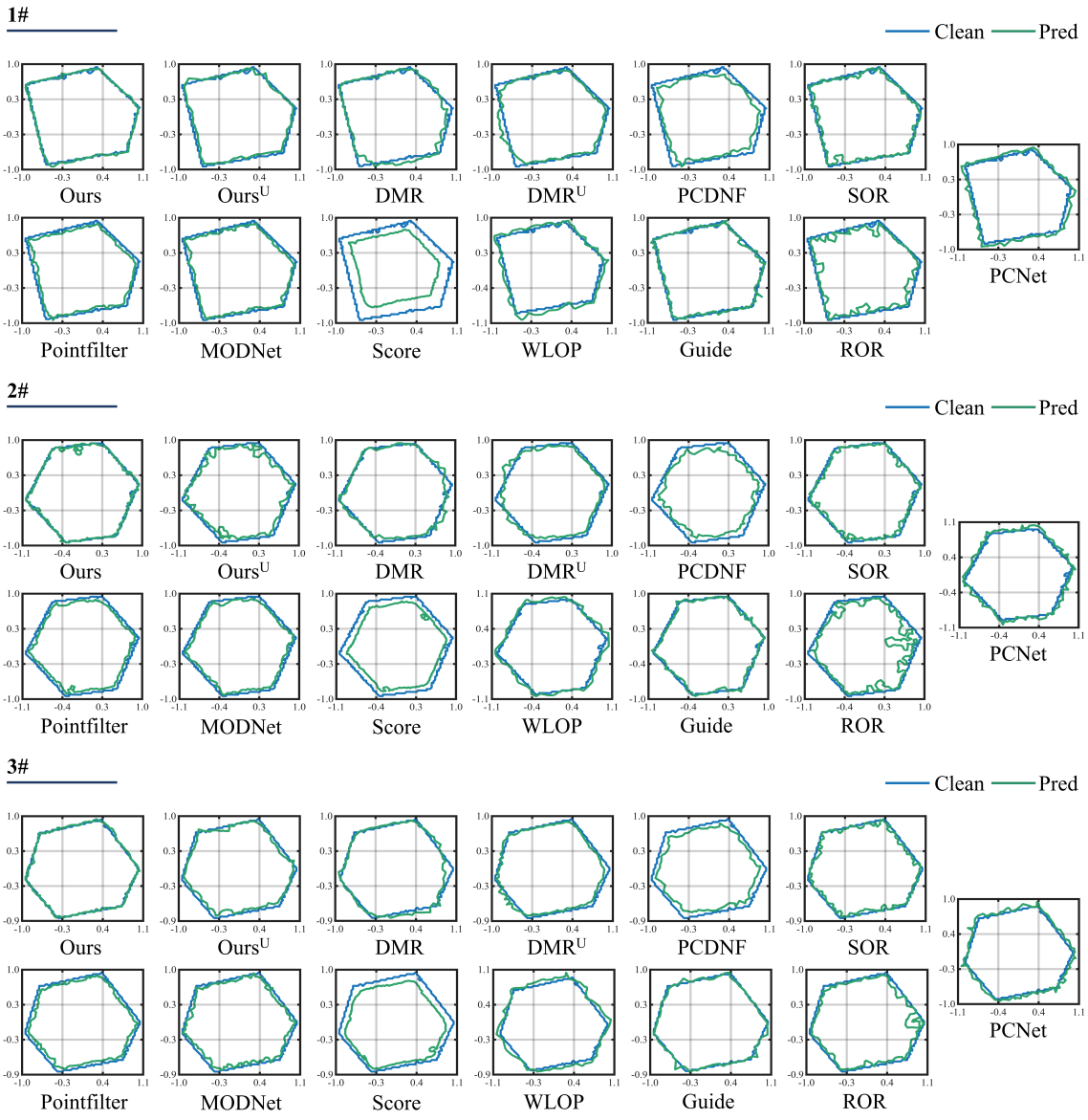

Figure 12: Comparison of the extracted 2D boundaries after denoising of baselines under typical noise condition.

Table 3: Boundary metric comparison.

Method	Area Ratio	2D Chamfer Distance
WLOP	1.0905 ± 0.0236	0.0477 ± 0.0057
Guide	1.0074 ± 0.0171	0.0282 ± 0.0024
SCORE	0.6911 ± 0.0914	0.1394 ± 0.0433
PCNet	1.1067 ± 0.0182	0.0549 ± 0.0050
MODNet	0.9046 ± 0.0102	0.0402 ± 0.0040
PCDNF	0.8288 ± 0.0145	0.0714 ± 0.0074
Pointfilter	0.8821 ± 0.0099	0.0492 ± 0.0044
DMR-U	0.9342 ± 0.0297	0.0475 ± 0.0073
DMR	0.9856 ± 0.0271	0.0401 ± 0.0076
SOR	0.9440 ± 0.0140	0.0294 ± 0.0058
ROR	0.8767 ± 0.0220	0.0484 ± 0.0108
OURS-U	0.9000 ± 0.0189	0.0431 ± 0.0063
OURS	1.0044 ± 0.0106	0.0190 ± 0.0017

Note: Values are reported as mean ± SD.

denoised point clouds relative to the clean counterparts. In addition, the unsupervised mode of SIREND-Net also exhibited clear competitive advantages in terms of the boundary-level metrics. The area enclosed by the denoised two-dimensional boundary decreased by only approximately 10%, and the Chamfer distance was lower than those obtained using methods such as DMR and PCDNF. These improvements were also attributed to the role of the N2P attention mechanism in preserving boundary-related geometric information. Overall, learning-based models tended to over-filter sonar terrain point clouds, leading to the loss of geometric information; consequently, they yielded smaller enclosed areas and greater geometric distortion, as indicated by increased CD. In contrast, conventional methods mainly suffered from residual noise, i.e., incomplete noise removal resulted in larger enclosed areas; however, in terms of boundary morphology, their CD values were better than those of most learning-based models. In comparison, the proposed SIREND-Net achieved the best overall balance between noise removal and geometric detail preservation. While reducing noise and thereby decreasing the Chamfer distance, it avoided the over-filtering effect that would otherwise have underestimated the predicted enclosed area.

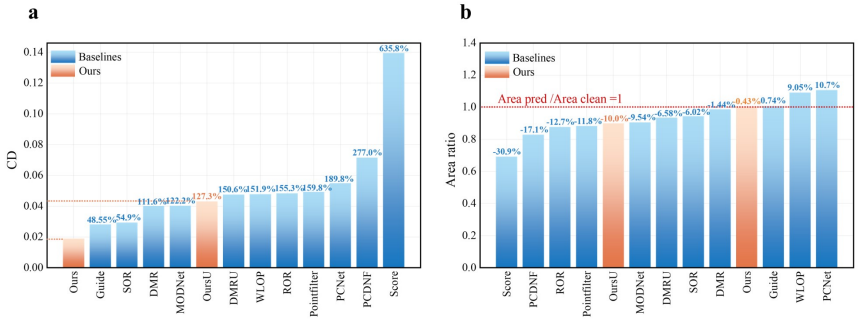

Figure 13: Quantitative comparison of extracted 2D boundaries after denoising under typical noise conditions. a. Chamfer distance between the extracted 2D boundary points from the clean and denoised point clouds. b. Enclosed area of the extracted 2D boundaries for the clean and denoised point clouds.

In this section, the effects of SOR and ROR, two filtering algorithms commonly used in industrial workflows, on the number of points retained after underwater sonar point-cloud

denoising were further investigated. The experiments were conducted using the USPCDenoise dataset under a representative noise condition of 4% Gaussian noise. the changes in the number of points during denoising under the representative noise condition were statistically compared, as shown in the figure below. Different degrees of point-cloud density reduction were observed for both SOR and ROR. After denoising, the number of points was reduced by an average of 12.9% for SOR and 46.2% for ROR. These results demonstrated that filtering-based denoising algorithms caused substantial density reduction and resulted in the loss of geometric structures and local features. Such information loss could irreversibly compromise subsequent defect detection and scour analysis.

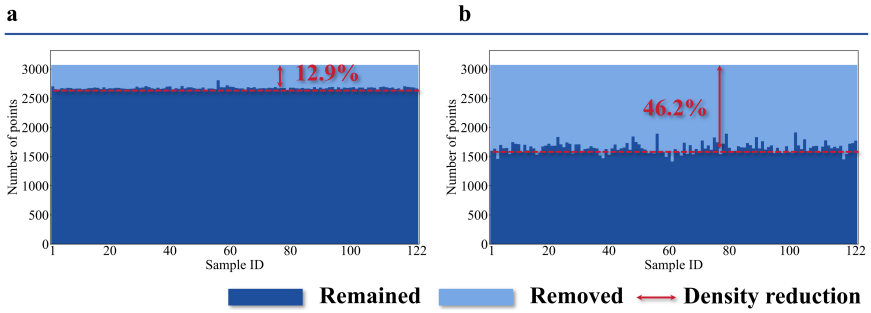

Figure 14: Effects of SOR and ROR on point-cloud density. a, Effect of SOR on point-cloud density. b, Effect of ROR on point-cloud density.

4.3 Quantitative evaluation of denoising for downstream tasks

To further evaluate the impact of point cloud denoising on downstream identification tasks, synthetic pile point clouds with defects and terrain point clouds with scour were constructed following the simulation and evaluation procedures in [1]. The identification errors of scour and defect regions before and after denoising were then compared.

For the pile data, each sample was generated as a cylindrical surface point cloud containing 8,000 points. The cylinder height and diameter were uniformly sampled from $[2.0, 4.0]$ and $[1.2, 2.4]$, respectively. The generated point clouds were normalized and mapped to the unfolded domain using the cylindrical unfolding method in [1]. For the terrain data, planar point clouds were directly generated. Thus, both data types were represented in a two-dimensional flattened domain for subsequent scour or defect simulation.

Local scour and surface defects were simulated using randomly generated Gaussian pits. Each sample contained 1 to 4 pits, with their centers restricted to the central region of the flattened plane to avoid boundary effects. The scale parameter σ and maximum depth A were uniformly sampled from $[0.06, 0.14]$ and $[0.20, 0.55]$, respectively. For the pile data, the unfolded point clouds were subsequently mapped back to the original cylindrical coordinate system using the corresponding fitted radius and unfolding information.

A total of 200 synthetic samples were generated, including 100 terrain point clouds with simulated scour and 100 pile point clouds with simulated defects. Gaussian noise at six levels from 1% to 6% was then added to construct test data under different noise conditions.

To quantify the effect of denoising on downstream identification, the terrain and pile point clouds before and after denoising were mapped onto a unified 20×20 grid map following Ref. [1] and compared with the corresponding ground truth. Mean absolute error (MAE) was used to quantify the discrepancy between the predicted grid maps and the ground truth. Fig. 15 presents the Grid MAE results before and after denoising under 3% to 5% noise levels.

Denoising consistently reduced the Grid MAE across all tested noise levels. For terrain point clouds with scour, the mean MAE decreased from approximately 0.05075 to 0.03822, corresponding to a 24.70% reduction. For pile point clouds with defects, it decreased from approximately

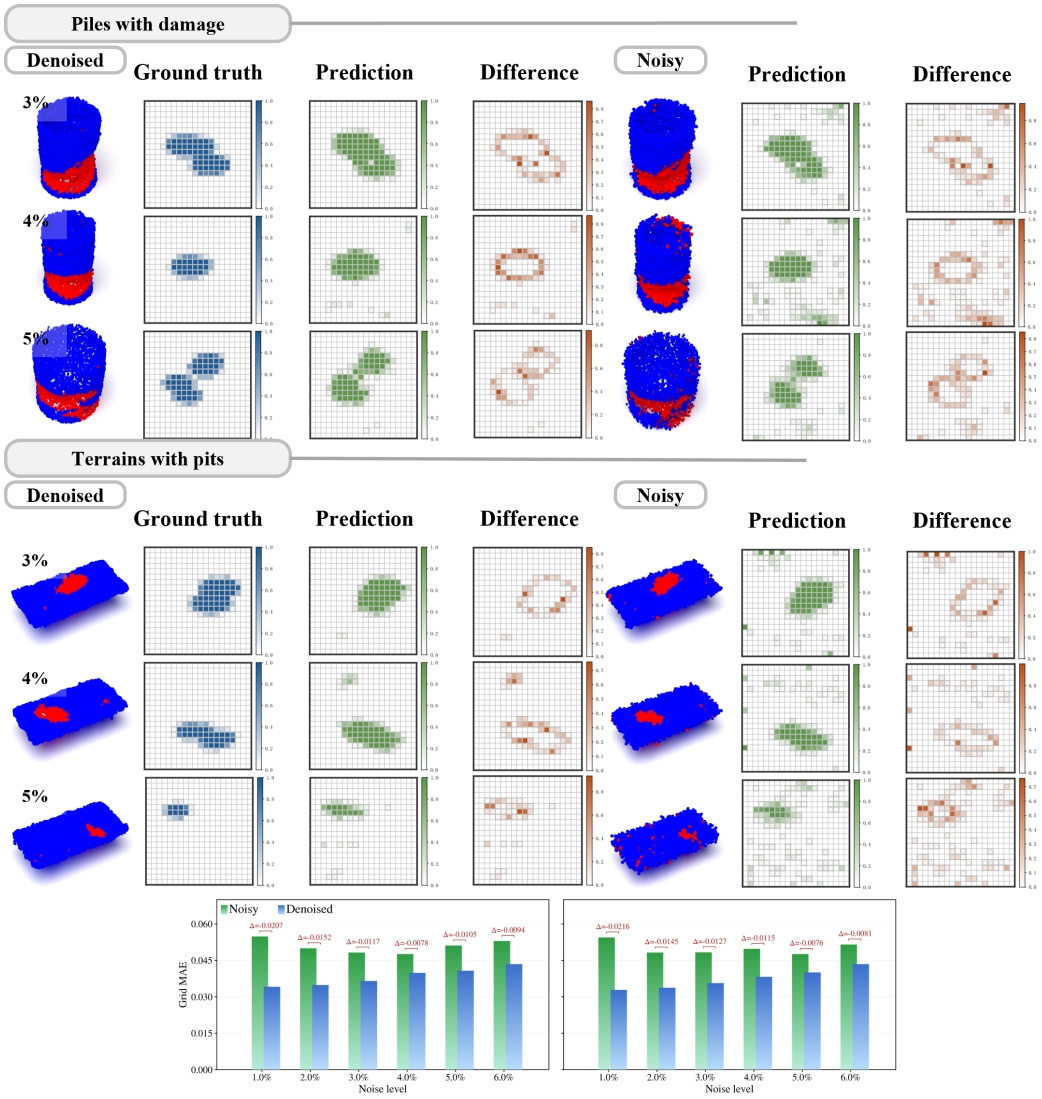

Figure 15: Quantitative evaluation of denoising for downstream tasks. a. Grid MAE results for pile point clouds with defects. b. Grid MAE results for terrain point clouds with scour.

0.04995 to 0.03728, corresponding to a 25.36% reduction. Similar trends were observed for both data types, indicating that SIREND-Net improves downstream region identification for different underwater targets. Meanwhile, the absolute MAE reduction generally decreased as the noise level increased, suggesting that residual noise had a greater influence on the spatial representation of scour and defect regions under severe noise conditions. Overall, these results demonstrate that the geometric improvements achieved by SIREND-Net can benefit subsequent identification and quantitative characterization of scour and defect regions.

5 Supervised-Unsupervised modes switching mechanism experiment

To enable an objective switch between SIREND-Net's supervised and unsupervised modes in practical engineering applications, experiments on the switching mechanism were conducted. Specifically, the underwater terrain sonar point cloud of USPCDenoise was used and initial Gaussian noise with a standard deviation of 1–6% was progressively added to the training set to emulate the initial degradation in dataset quality in the engineering scene. On this basis, the experimental settings of the supervised mode performance evaluation were repeated. i.e., Gaussian noise with a standard deviation of 1–6% was again added to the normalized noisy training set to emulate supervised mode training under deteriorating dataset quality.

Finally, the model was evaluated on the test set added with Gaussian noise with a standard deviation of 1–6%, and the results were compared with the results of the unsupervised mode in the performance experiments, as shown in Fig. 16. As data quality deteriorated, the performance of supervised training conducted by additionally injecting simulated noise on top of the degraded training data also degraded progressively. When the initial Gaussian noise added to the training set reached approximately 3%, the performance of the supervised and unsupervised modes became comparable.

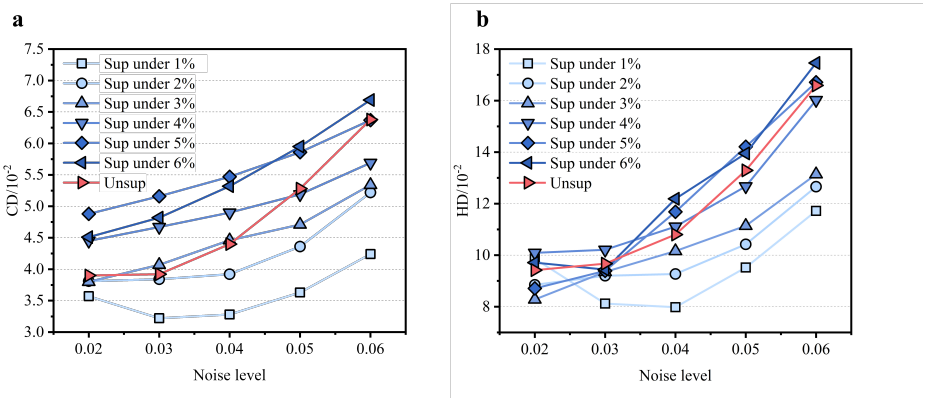

Figure 16: Results of the experiment on the supervised/unsupervised mode switching mechanism. a Chamfer distance results. b Hausdorff distance results.

Furthermore, it was inferred that when the training set contained initial Gaussian noise exceeding 3%, the unsupervised mode of SIREND-Net outperformed the supervised mode. Therefore, the relationship between the initial noise level in the noisy training set and the corresponding equivalent thickness was calculated, and the nominal noise level was subsequently derived, as shown in Fig. 17. Based on the fitted relationship, the nominal noise level corresponding to 3% initial Gaussian noise was estimated to be 2.3%.

Accordingly, for practical deployment of SIREND-Net, it was recommended that the nominal noise intensity first be evaluated from the equivalent thickness to assess dataset quality. When the nominal noise intensity exceeded 2.3%, the unsupervised mode of SIREND-Net was recommended

as the preferred option to mitigate the degradation in supervised learning performance caused by reduced data quality.

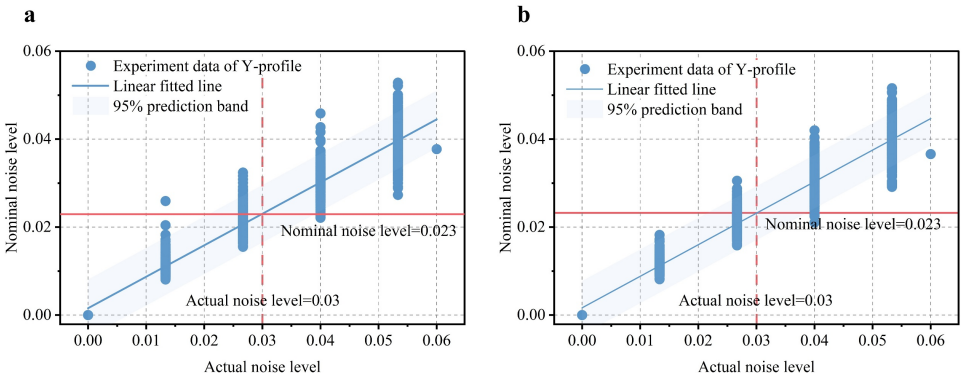

Figure 17: Estimation of the nominal noise level for switching between supervised and unsupervised modes. a The nominal noise level was estimated from the central X-profile. b The nominal noise level was estimated from the central Y-profile.

6 Expanded analysis of model generalization performance

In this section, we report the test results of all baselines and SIREND-Net under Uniform noise (5%, 10%, and 15%), Discrete noise (10%, 12%, and 14%), and Laplacian noise (2%, 3%, and 4%) to evaluate their generalization ability. The generalization results are shown in Fig. 18. Under 15% Uniform noise, the CD and HD values of the competing methods were, on average, 67.44% and 49.14% higher, respectively, than those of SIREND-Net, and under 4% Laplacian noise, they were 61.81% and 45.31% higher, respectively. For Discrete noise, performance remained competitive, though CD degraded slightly relative to other noise types. This is likely due to the anisotropy and discreteness of Discrete noise in the spatial coordinate system, which differs substantially from the near-isotropic distributions of Gaussian and Uniform noise that resemble small-scale fuzzy noise. Nevertheless, even under 14% Discrete noise, the competing methods showed, on average, 6.88% higher CD and 51.62% higher HD than SIREND-Net. These results confirm both the competitive denoising performance and the strong generalization ability of the proposed method. The results are shown in Fig. 18. Table 4-8 presents the mean and standard deviation of each algorithm under different noise levels. The reported means and standard deviations are calculated across all test samples within a single experimental run to characterize the variability in denoising performance of each method.

Furthermore, a train–test cross-setting experiment was conducted using the underwater mapping target data in USPCDenoise. The cross settings were configured as follows. All learning-based models were first trained with t-distributed noise (noise level 1–6%, degrees of freedom 5–10) and were then tested under (i) Gaussian noise with a noise level of 3–5% (T–G) and (ii) t-distributed noise with a noise level of 3–5% and degrees of freedom 5–10 (T–T). In parallel, all models were trained with Gaussian noise (noise level 1–6%) and were then tested under (i) Gaussian noise with a noise level of 3–5% (G–G) and (ii) t-distributed noise with a noise level of 3–5% and degrees of freedom 5–10 (G–T). These cross-setting experiments revealed how training with different simulated noise distributions affected the generalization performance of network models on underwater sonar point clouds. The results were summarized in Table 9.

As shown in Table 9, the CD and HD obtained under the G–T setting were better than those under the T–T setting for most models. This outcome was likely attributable to the increased number of extreme outliers generated when t-distributed noise with low degrees of freedom was used for simulation, which could have impaired training stability across baselines and, consequently, degraded denoising performance. In other words, for most models, denoising

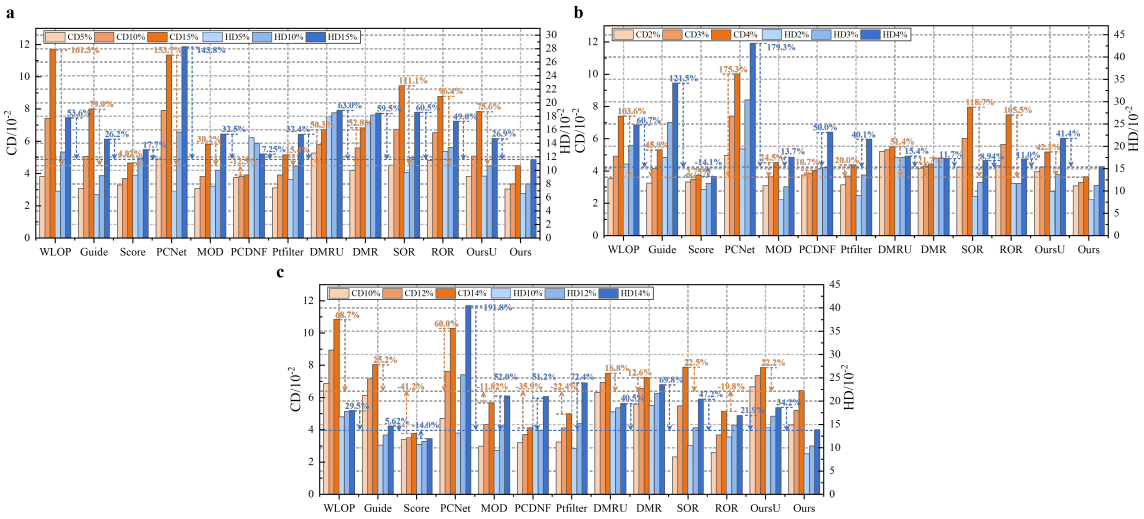

Figure 18: Quantitative results of numerical experiments. OursU and DMRU denote the unsupervised learning modes of SIREND-Net and DMR, respectively. a Generalization results under Uniform noise. b Generalization results under Laplace noise. c Generalization results under Discrete noise.

Table 4: Numerical performance under Gaussian noise.

Method	0.02		0.03		0.04		0.05		0.06	
	CD ↓	HD ↓	CD ↓	HD ↓	CD ↓	HD ↓	CD ↓	HD ↓	CD ↓	HD ↓
WLOP	3.54 ± 0.13	8.40 ± 1.55	4.62 ± 0.16	14.88 ± 2.29	6.57 ± 0.19	17.69 ± 2.13	8.88 ± 0.25	19.77 ± 2.19	11.17 ± 0.36	23.14 ± 1.99
Guide	3.00 ± 0.07	6.26 ± 0.55	3.43 ± 0.09	9.51 ± 1.08	4.23 ± 0.13	14.56 ± 1.49	5.28 ± 0.17	18.38 ± 2.33	6.44 ± 0.22	22.35 ± 2.94
SCORE	3.24 ± 0.30	9.26 ± 4.33	3.36 ± 0.31	10.43 ± 4.42	3.49 ± 0.28	11.35 ± 4.27	3.70 ± 0.29	12.13 ± 4.30	3.99 ± 0.25	13.05 ± 4.24
PCNet	4.42 ± 0.40	8.83 ± 5.92	6.20 ± 0.44	15.52 ± 5.81	8.09 ± 0.53	22.59 ± 5.99	10.11 ± 0.58	31.05 ± 5.94	12.30 ± 0.52	40.30 ± 5.65
MODNet	3.00 ± 0.19	7.47 ± 2.46	3.33 ± 0.18	8.76 ± 2.34	3.89 ± 0.14	10.36 ± 2.29	4.82 ± 0.15	14.24 ± 2.23	6.23 ± 0.21	20.13 ± 3.21
PCDNF	3.73 ± 0.22	14.96 ± 1.66	3.80 ± 0.24	14.63 ± 1.84	3.83 ± 0.25	13.96 ± 1.98	3.90 ± 0.26	14.05 ± 2.36	4.14 ± 0.25	21.16 ± 3.91
Pointfilter	3.03 ± 0.19	8.35 ± 1.48	3.41 ± 0.19	9.76 ± 1.72	3.91 ± 0.17	11.13 ± 1.67	4.51 ± 0.15	15.59 ± 2.66	5.64 ± 0.18	22.80 ± 2.93
DMR-U	5.09 ± 0.40	17.47 ± 5.45	5.20 ± 0.30	17.45 ± 4.86	5.32 ± 0.24	17.88 ± 4.68	5.47 ± 0.22	17.96 ± 4.53	6.01 ± 0.21	18.31 ± 4.32
DMR	4.14 ± 0.34	17.26 ± 5.08	4.27 ± 0.27	17.32 ± 4.79	4.41 ± 0.21	17.20 ± 4.64	4.60 ± 0.19	17.82 ± 4.18	4.81 ± 0.19	17.43 ± 4.04
SOR	4.42 ± 0.10	12.59 ± 4.88	5.12 ± 0.10	12.54 ± 4.68	5.70 ± 0.11	12.50 ± 3.87	6.42 ± 0.12	12.92 ± 3.81	7.10 ± 0.14	13.24 ± 2.93
ROR	4.48 ± 0.15	10.28 ± 5.12	5.10 ± 0.13	10.60 ± 5.15	5.78 ± 0.12	10.85 ± 4.36	6.52 ± 0.13	11.83 ± 4.51	7.22 ± 0.14	13.43 ± 4.34
OURS-U	3.90 ± 0.10	9.42 ± 3.80	3.92 ± 0.10	9.67 ± 2.16	4.40 ± 0.11	10.80 ± 1.80	5.28 ± 0.10	13.29 ± 2.32	6.38 ± 0.12	16.59 ± 2.47
OURS	3.02 ± 0.11	6.55 ± 1.63	3.12 ± 0.09	6.99 ± 1.24	3.26 ± 0.10	8.24 ± 1.76	3.49 ± 0.09	9.74 ± 1.84	3.79 ± 0.10	11.52 ± 2.15

Note: Values are reported as mean ± SD. The CD and HD values were on the order of 10^{-2}.

training with Gaussian noise simulation yielded better generalization under the T setting than training directly with t-distributed noise. This finding suggested that, for sonar point-cloud denoising, Gaussian-noise simulation was more effective than t-distributed noise simulation in practical engineering applications because it more closely reproduced the dominant morphological characteristics of real sonar point-cloud noise.

7 Expanded on-site sub-experiment: Denoising sonar point clouds in environments with dense aquatic organisms and suspended particulates

To validate the effectiveness of the proposed network under the unsupervised setting in removing potential outlier noise induced by aquatic organisms and suspended particulates in complex underwater environments, an additional extended experiment was conducted based on field measurements.

Table 5: Numerical performance under Laplacian noise.

Method	0.02		0.03		0.04	
	CD ↓	HD ↓	CD ↓	HD ↓	CD ↓	HD ↓
WLOP	3.52 ± 0.15	15.99 ± 3.26	4.92 ± 0.31	20.20 ± 2.51	7.41 ± 0.53	24.80 ± 3.33
Guide	3.26 ± 0.15	17.43 ± 3.37	4.15 ± 0.14	25.33 ± 3.93	5.31 ± 0.13	34.17 ± 5.45
SCORE	3.32 ± 0.30	10.28 ± 4.35	3.50 ± 0.29	11.71 ± 4.16	3.74 ± 0.29	13.25 ± 4.26
PCNet	4.96 ± 0.28	19.38 ± 3.54	7.41 ± 0.30	30.38 ± 3.97	10.02 ± 0.30	43.10 ± 4.37
MODNet	3.09 ± 0.20	8.02 ± 2.13	3.64 ± 0.17	10.91 ± 2.40	4.53 ± 0.15	17.55 ± 3.19
PCDNF	3.75 ± 0.22	14.92 ± 1.22	3.86 ± 0.24	15.20 ± 2.32	4.03 ± 0.25	23.16 ± 5.44
Pointfilter	3.14 ± 0.20	8.91 ± 1.25	3.68 ± 0.18	13.59 ± 3.11	4.37 ± 0.15	21.62 ± 3.41
DMR-U	5.20 ± 0.32	17.46 ± 4.35	5.31 ± 0.24	17.54 ± 5.79	5.51 ± 0.22	17.81 ± 4.76
DMR	4.18 ± 0.30	17.32 ± 4.86	4.31 ± 0.23	17.35 ± 4.05	4.43 ± 0.19	17.23 ± 5.10
SOR	4.24 ± 0.10	8.78 ± 5.03	6.02 ± 0.12	11.89 ± 4.08	7.96 ± 0.17	16.81 ± 3.57
ROR	4.32 ± 0.13	11.60 ± 4.87	5.66 ± 0.12	11.65 ± 3.41	7.48 ± 0.17	17.13 ± 2.79
OURS-U	3.96 ± 0.10	9.84 ± 2.74	4.26 ± 0.10	13.67 ± 3.98	5.18 ± 0.11	21.81 ± 4.87
OURS	3.08 ± 0.10	8.01 ± 2.35	3.29 ± 0.08	11.22 ± 3.39	3.64 ± 0.09	15.43 ± 3.18

Note: Values are reported as mean ± SD. The CD and HD values were on the order of 10^{-2}.

Table 6: Numerical performance under Discrete noise.

Method	0.10		0.12		0.14	
	CD ↓	HD ↓	CD ↓	HD ↓	CD ↓	HD ↓
WLOP	6.86 ± 0.22	16.56 ± 0.82	8.94 ± 0.31	17.77 ± 1.33	10.85 ± 0.52	17.96 ± 1.57
Guide	6.13 ± 0.18	10.50 ± 0.15	7.17 ± 0.18	12.70 ± 0.20	8.05 ± 0.19	14.65 ± 0.15
SCORE	3.39 ± 0.32	10.67 ± 4.04	3.50 ± 0.30	11.35 ± 4.08	3.78 ± 0.24	11.93 ± 4.06
PCNet	4.71 ± 0.31	13.09 ± 3.98	7.61 ± 0.33	25.66 ± 3.97	10.29 ± 0.37	40.47 ± 4.01
MODNet	2.99 ± 0.22	9.33 ± 2.25	4.33 ± 0.16	14.68 ± 2.78	5.67 ± 0.18	21.08 ± 2.04
PCDNF	3.21 ± 0.28	14.68 ± 1.65	3.70 ± 0.25	13.65 ± 2.07	4.12 ± 0.26	20.97 ± 1.49
Pointfilter	3.24 ± 0.17	9.75 ± 1.75	4.11 ± 0.16	15.17 ± 2.69	4.99 ± 0.16	23.91 ± 2.96
DMR-U	6.31 ± 0.33	17.64 ± 4.14	6.93 ± 0.36	18.53 ± 4.15	7.51 ± 0.42	19.49 ± 3.68
DMR	5.59 ± 0.42	19.02 ± 4.45	6.56 ± 0.49	21.62 ± 4.87	7.24 ± 0.55	23.55 ± 3.99
SOR	2.32 ± 0.22	10.45 ± 3.92	5.48 ± 0.25	14.29 ± 3.05	7.88 ± 0.23	20.41 ± 2.84
ROR	2.60 ± 0.17	12.25 ± 4.91	3.68 ± 0.24	14.82 ± 4.73	5.16 ± 0.50	16.90 ± 3.77
OURS-U	6.66 ± 0.16	14.34 ± 3.35	7.35 ± 0.22	16.76 ± 2.34	7.86 ± 0.21	18.61 ± 2.98
OURS	4.30 ± 0.16	8.64 ± 2.18	5.21 ± 0.40	10.37 ± 1.95	6.43 ± 0.52	13.87 ± 2.38

Note: Values are reported as mean ± SD. The CD and HD values were on the order of 10^{-2}.

7.1 Engineering background

The extended experiment was conducted at a bridge site in Anhui Province, China, and the field measurement environment was shown in Fig. 19. The water body in the surveyed area was turbid and contained abundant suspended particles. Moreover, eutrophication was observed, and substantial aquatic vegetation was present. During instrument retrieval, as shown in Fig. 20, extensive aquatic weeds were observed to be entangled around the sonar. Accordingly, typical data from this area were selected to illustrate the potential noise characteristics introduced into sonar point clouds by aquatic organisms and suspended particulates.

Representative terrain mapping data from this area were visualized using the built-in acquisition software of the Coda Octopus Echoscope4G C500 sonar, as shown in Fig. 20. As shown, the regions highlighted by the red circles corresponded to noise points that deviated from the terrain-mapping point cloud, and their density was substantially lower than that of the main terrain returns. For such outlier noise induced by aquatic vegetation or suspended

Table 7: Numerical performance under Uniform noise.

Method	0.05		0.10		0.15	
	CD ↓	HD ↓	CD ↓	HD ↓	CD ↓	HD ↓
WLOP	3.81 ± 0.15	6.89 ± 0.69	7.42 ± 0.17	12.72 ± 1.00	11.68 ± 0.20	17.80 ± 1.35
Guide	3.08 ± 0.14	6.36 ± 0.82	5.06 ± 0.20	9.22 ± 0.30	8.00 ± 0.27	14.63 ± 0.24
SCORE	3.28 ± 0.30	9.24 ± 4.29	3.69 ± 0.28	11.81 ± 4.00	4.65 ± 0.18	13.07 ± 3.96
PCNet	4.90 ± 0.37	6.91 ± 3.93	7.91 ± 0.39	15.67 ± 4.09	11.34 ± 0.44	28.26 ± 4.30
MODNet	3.07 ± 0.19	7.64 ± 2.42	3.81 ± 0.16	10.01 ± 3.95	5.82 ± 0.19	15.36 ± 2.98
PCDNF	3.75 ± 0.23	14.80 ± 1.14	3.81 ± 0.25	14.00 ± 1.73	3.91 ± 0.26	12.43 ± 2.22
Pointfilter	3.11 ± 0.20	8.65 ± 1.99	3.91 ± 0.18	10.64 ± 1.62	5.16 ± 0.16	15.34 ± 3.46
DMR-U	5.27 ± 0.37	17.88 ± 5.26	5.79 ± 0.23	18.51 ± 4.60	6.72 ± 0.26	18.89 ± 4.48
DMR	4.18 ± 0.29	17.01 ± 4.76	5.58 ± 0.19	18.19 ± 4.45	6.83 ± 0.21	18.49 ± 4.69
SOR	4.52 ± 0.09	9.67 ± 4.90	6.74 ± 0.14	11.95 ± 4.06	9.44 ± 0.20	18.60 ± 2.63
ROR	4.82 ± 0.13	12.78 ± 4.82	6.52 ± 0.14	13.40 ± 4.76	8.78 ± 0.22	17.27 ± 3.21
OURS-U	3.82 ± 0.10	9.13 ± 2.21	5.08 ± 0.11	10.65 ± 1.98	7.85 ± 0.12	14.71 ± 2.17
OURS	3.04 ± 0.09	6.53 ± 1.29	3.35 ± 0.09	8.00 ± 1.12	4.47 ± 0.11	11.59 ± 1.79

Note: Values are reported as mean ± SD. The CD and HD values were on the order of 10^{-2}.

Table 8: Numerical performance under t-distribution noise.

Method	0.03		0.04		0.05	
	CD ↓	HD ↓	CD ↓	HD ↓	CD ↓	HD ↓
WLOP	6.01 ± 1.30	20.40 ± 4.61	9.42 ± 1.39	26.70 ± 8.79	12.44 ± 1.20	31.06 ± 7.62
Guide	4.26 ± 1.49	19.14 ± 5.43	5.72 ± 1.39	26.68 ± 8.90	7.47 ± 1.47	32.21 ± 7.61
SCORE	3.98 ± 0.47	19.66 ± 4.72	4.93 ± 0.48	26.16 ± 4.81	5.74 ± 0.49	31.17 ± 4.74
PCNet	8.62 ± 0.50	18.84 ± 4.36	8.89 ± 0.48	24.98 ± 4.11	10.46 ± 0.50	36.40 ± 4.28
MODNet	3.34 ± 0.18	8.94 ± 2.21	3.82 ± 0.17	10.86 ± 3.92	4.67 ± 0.15	15.55 ± 2.65
PCDNF	3.81 ± 0.24	15.60 ± 7.05	3.87 ± 0.25	17.95 ± 10.98	4.37 ± 0.26	23.95 ± 10.88
Pointfilter	3.41 ± 0.18	9.75 ± 1.70	4.04 ± 0.18	14.27 ± 2.48	4.64 ± 0.16	18.33 ± 3.52
DMR-U	4.75 ± 0.25	17.73 ± 14.91	5.30 ± 0.22	22.53 ± 28.39	6.29 ± 0.24	24.72 ± 20.50
DMR	4.21 ± 0.24	16.30 ± 8.86	4.45 ± 0.19	16.61 ± 11.67	4.86 ± 0.20	18.01 ± 10.70
SOR	5.28 ± 0.11	9.66 ± 4.26	6.64 ± 0.14	13.62 ± 3.19	8.18 ± 0.17	18.44 ± 2.80
ROR	5.16 ± 0.12	11.49 ± 4.70	6.24 ± 0.14	13.19 ± 4.12	7.56 ± 0.19	18.04 ± 3.78
OURS-U	4.51 ± 0.10	12.11 ± 8.54	5.49 ± 0.11	17.72 ± 10.54	6.09 ± 0.14	21.96 ± 10.34
OURS	3.30 ± 0.10	9.59 ± 8.23	3.71 ± 0.11	13.93 ± 10.87	4.58 ± 0.12	17.80 ± 10.16

Note: Values are reported as mean ± SD. The CD and HD values were on the order of 10^{-2}.

particulates, the proposed SIREND-Net exploited the N2P attention mechanism to model the relevance between each point and its neighborhood. Specifically, keys were constructed from within-neighborhood differences to enhance geometric saliency. Under the unsupervised setting, this design enabled the model to focus on sparse outliers, e.g., those generated by aquatic weeds and suspended particles, and to remove them in the first stage, thereby preventing erroneous convergence.

7.2 Denoising results and visualization of N2P

To validate the denoising performance of SIREND-Net under the unsupervised setting in real-world environments with high turbidity and dense aquatic vegetation, representative samples were selected from this survey. The unsupervised model trained in the performance experiments was applied to denoise the field data, and the point clouds before and after denoising were shown in Fig. 21. As shown, returns attributable to suspended particles, as well as locally

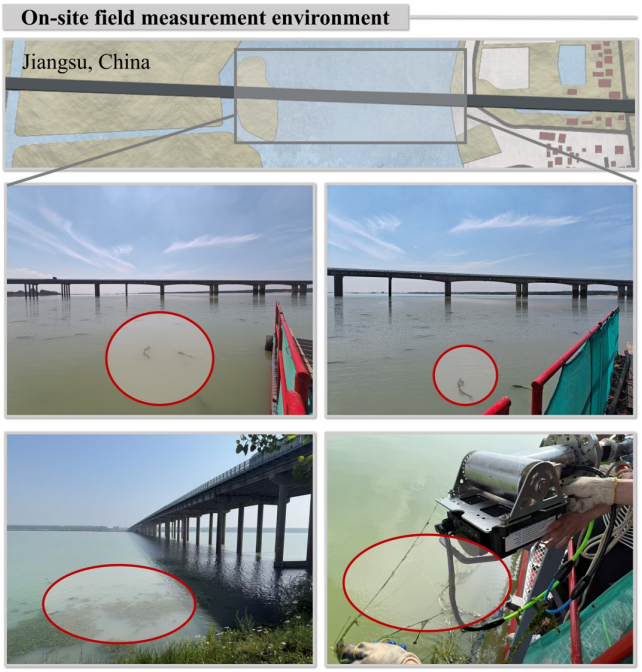

Figure 19: Photographs of aquatic organisms and vegetation observed in the surveyed waters for the representative field data.

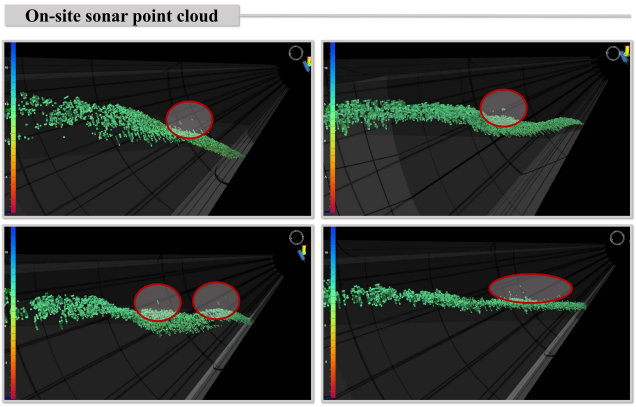

Figure 20: Sparse outlier noise deviating from the riverbed terrain, induced by aquatic vegetation or suspended particulates.

Table 9: Performance generalization cross-setting experiments. The arrows indicate that lower values are better.

Bsl.	Train–Test Model	Noise level					
		3%		4%		5%	
		CD ↓	HD ↓	CD ↓	HD ↓	CD ↓	HD ↓
Ours	G-G	**3.12**	**6.99**	**3.26**	**8.24**	**3.49**	**9.74**
	G-T	3.30	9.59	3.71	13.93	4.58	17.80
	T-T	3.47	10.17	4.70	15.27	5.58	20.16
	T-G	3.19	7.01	3.70	8.36	4.57	10.50
OursU	G-G	3.92	9.67	4.40	10.80	5.28	13.29
	G-T	4.51	12.11	5.49	17.72	6.09	21.96
	T-T	4.54	23.37	5.76	21.93	6.16	24.63
	T-G	4.48	22.71	4.68	22.19	5.47	20.40
DMR	G-G	4.27	17.32	4.41	17.20	4.60	17.82
	G-T	4.21	16.30	4.45	16.61	4.86	18.01
	T-T	4.46	19.04	4.68	18.72	5.08	18.48
	T-G	4.40	19.16	4.52	18.97	4.66	18.58
DMRU	G-G	5.20	17.45	5.32	17.88	5.47	17.96
	G-T	4.75	17.73	5.30	22.53	6.29	24.72
	T-T	5.60	17.85	6.03	18.62	6.52	19.59
	T-G	5.47	17.60	5.73	18.17	6.00	18.49
PCNet	G-G	6.20	15.52	8.09	22.59	10.11	31.05
	G-T	8.62	18.84	8.89	24.98	10.46	36.40
	T-T	9.23	19.10	10.32	25.85	11.07	37.72
	T-G	8.64	19.04	9.02	24.87	10.99	34.26
Pointfilter	G-G	3.41	9.76	3.91	11.13	4.51	15.59
	G-T	3.41	9.75	4.04	14.27	4.64	18.33
	T-T	3.42	9.78	4.08	14.30	4.67	18.67
	T-G	3.28	9.22	3.52	10.11	3.79	10.49
PCDNF	G-G	3.80	14.63	3.83	13.96	3.90	14.05
	G-T	3.81	15.60	3.87	17.95	4.37	23.95
	T-T	3.88	15.53	3.92	18.03	4.51	24.12
	T-G	3.84	14.86	3.88	14.70	3.84	14.19
MODNET	G-G	3.33	8.76	3.89	10.36	4.82	14.24
	G-T	3.34	8.94	3.82	10.86	4.67	16.55
	T-T	3.35	8.97	3.83	10.92	4.66	16.44
	T-G	3.22	8.46	3.44	9.34	3.75	9.80
Score	G-G	3.36	10.43	3.49	11.35	3.70	12.13
	G-T	3.98	19.66	4.93	26.16	5.74	31.17
	T-T	4.13	19.73	5.06	26.36	5.88	31.72
	T-G	3.80	13.26	4.40	16.16	4.98	18.64

protruding points that were likely caused by aquatic vegetation, were either completely removed or effectively suppressed. These results indicated that the proposed network, even in unsupervised mode, effectively mitigated real-world noise plausibly induced by aquatic weeds or suspended particulates.

Furthermore, the N2P attention mechanism was visualized based on the above case to analyze how SIREND-Net avoided erroneous convergence, as shown in Fig. 22. As indicated

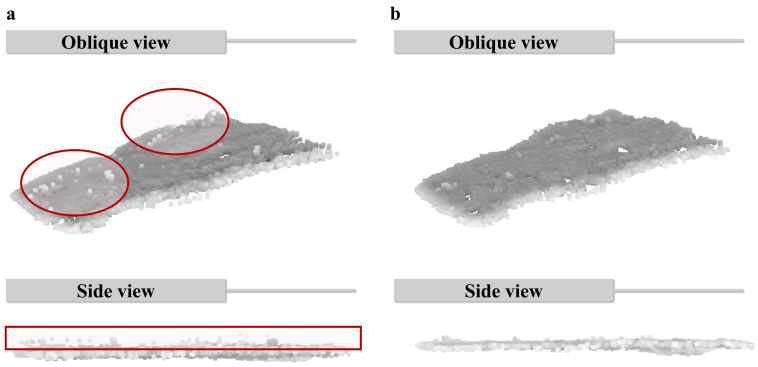

Figure 21: Unsupervised denoising results of SIREND-Net on representative field sonar point clouds acquired in a turbid environment with abundant aquatic vegetation. a. Before denoising; b. After denoising.

by the normalized correlation map that encoded feature differences between each point and its local neighborhood, the N2P attention primarily focused on two regions of the denoising patch: (i) boundary areas and (ii) potential outlier noise. This behavior suppressed outliers while preventing the model from converging to noise clusters plausibly caused by aquatic weeds or suspended particulates. Meanwhile, edge details were preserved, which reduced the risk of over-shrinkage when denoising numerous sub-patches in large-scale point clouds in practical engineering scenarios, thereby mitigating terrain cracks or gaps that could otherwise interfere with downstream tasks.

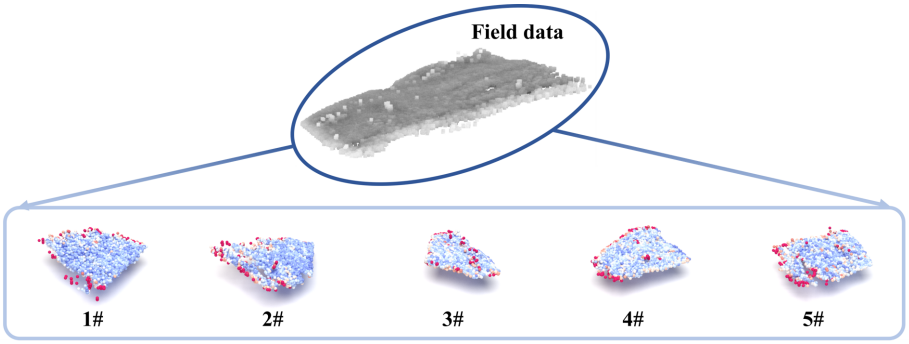

Figure 22: Visualization of N2P attention on representative noisy field sonar point cloud acquired in a turbid environment with dense aquatic vegetation.

8 Supplemental material results for the on-site performance and generalization comparison experiments

In this section, the remaining data (samples 2#–6#) used in the on-site performance and generalizability comparison experiments described in the main text were qualitatively visualized and analyzed, as shown in Figs. 23-27 and Table 10.

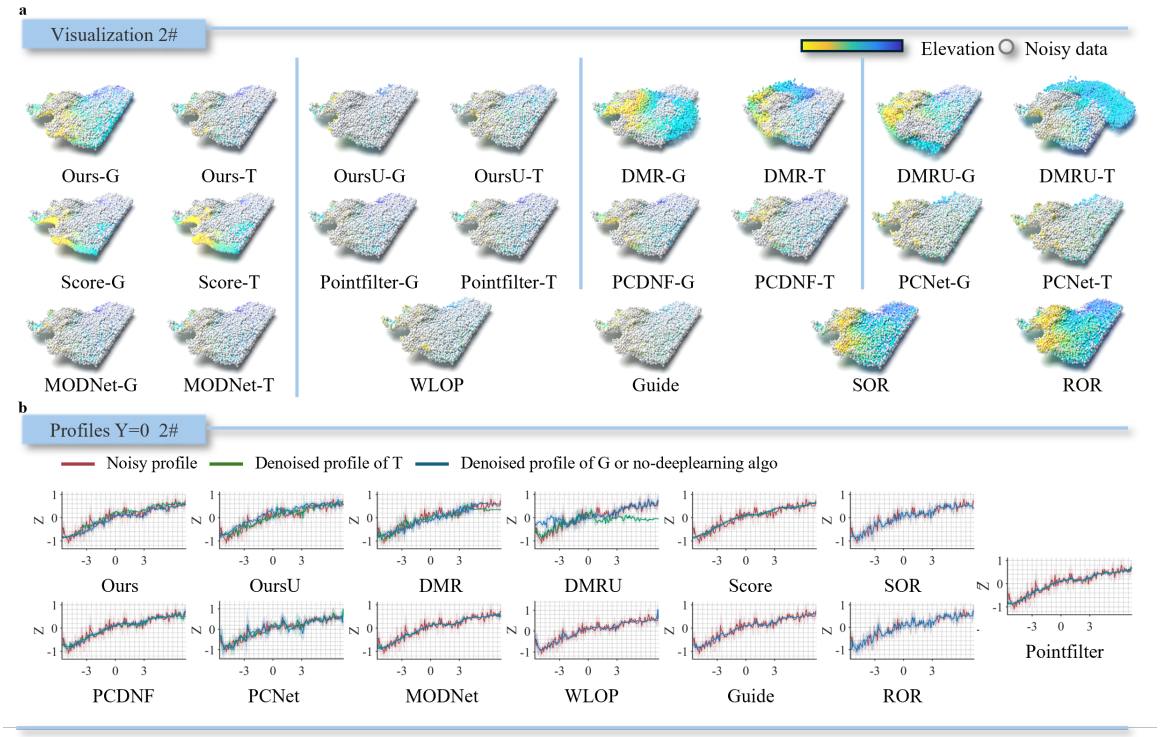

Figure 23: On-site performance and generalization comparison experiments (2#): a. Qualitative visualization on a representative sample; b. Comparison of the central Y-profile for the representative sample.

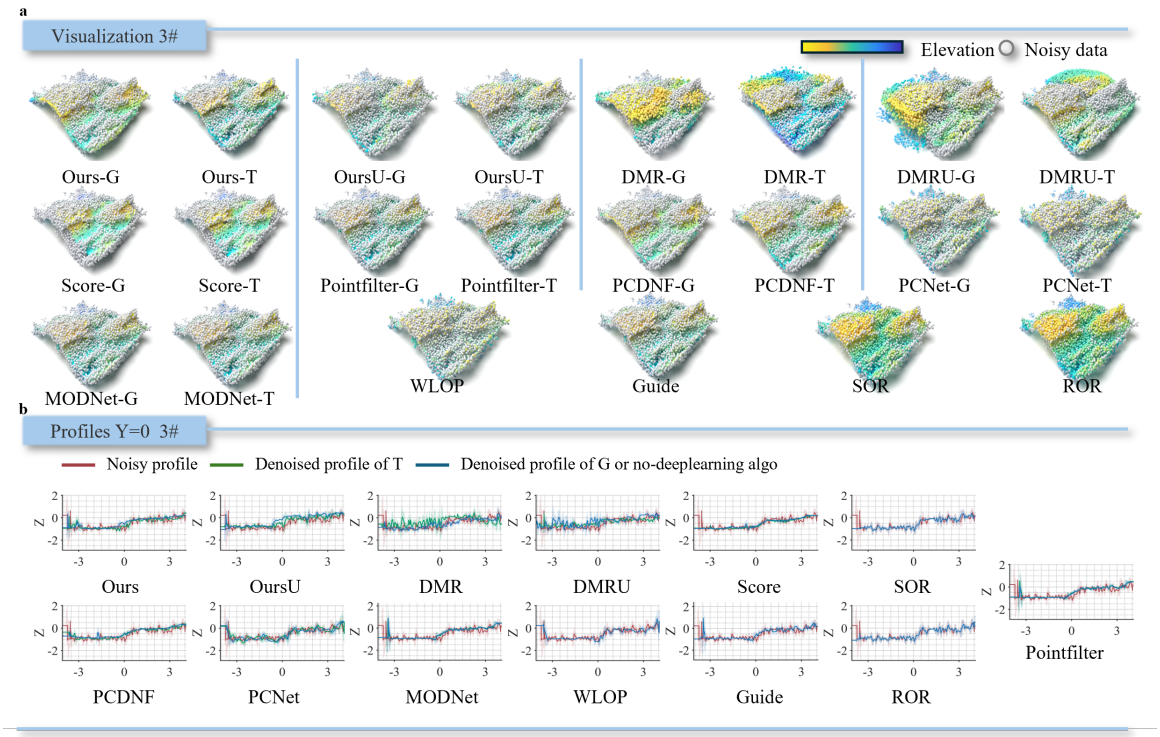

Figure 24: On-site performance and generalization comparison experiments (3#): a. Qualitative visualization on a representative sample; b. Comparison of the central Y-profile for the representative sample.

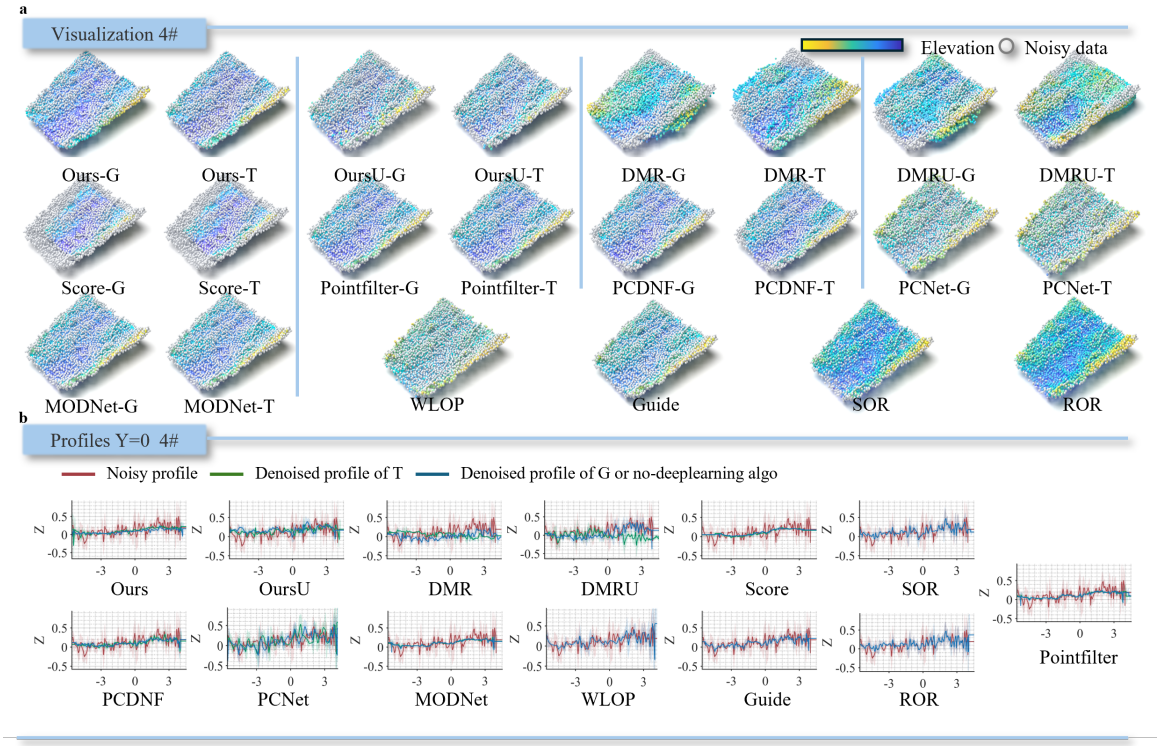

Figure 25: On-site performance and generalization comparison experiments (4#): a. Qualitative visualization on a representative sample; b. Comparison of the central Y-profile for the representative sample.

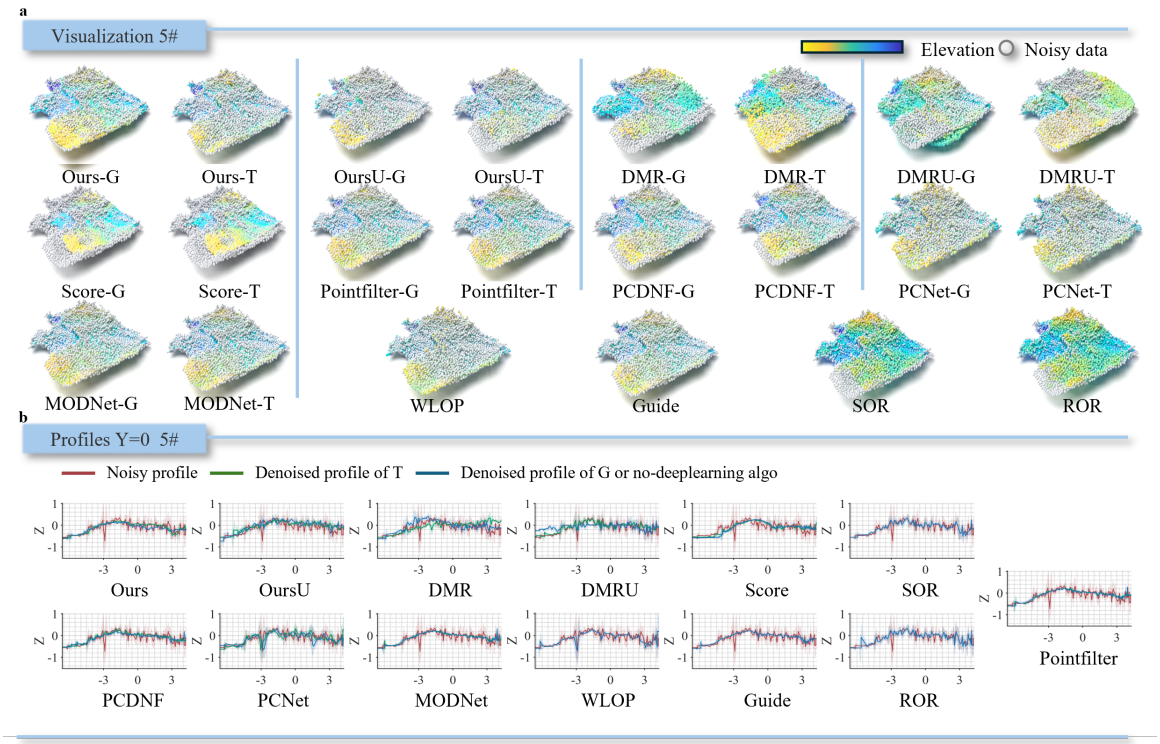

Figure 26: On-site performance and generalization comparison experiments (5#): a. Qualitative visualization on a representative sample; b. Comparison of the central Y-profile for the representative sample.

Table 10: Cross-section thickness comparison.

Method	X=0 Thickness_T	X=0 Thickness_G	Y=0 Thickness_T	Y=0 Thickness_G
WLOP	N/A	0.0791 ± 0.0241	N/A	0.0952 ± 0.0402
Guide	N/A	0.0667 ± 0.0272	N/A	0.0723 ± 0.0425
SCORE	0.0357 ± 0.0154	0.0327 ± 0.0143	0.0349 ± 0.0164	0.0283 ± 0.0146
PCNet	0.0939 ± 0.0243	0.0918 ± 0.0304	0.1090 ± 0.0267	0.1071 ± 0.0367
MODNet	0.0440 ± 0.0190	0.0477 ± 0.0207	0.0416 ± 0.0252	0.0455 ± 0.0281
PCDNF	0.0580 ± 0.0188	0.0437 ± 0.0059	0.0520 ± 0.0141	0.0368 ± 0.0087
Pointfilter	0.0514 ± 0.0193	0.0471 ± 0.0177	0.0461 ± 0.0260	0.0418 ± 0.0250
DMR-U	0.1131 ± 0.0516	0.0763 ± 0.0189	0.1204 ± 0.0668	0.0867 ± 0.0146
DMR	0.0970 ± 0.0494	0.0785 ± 0.0308	0.1019 ± 0.0769	0.0910 ± 0.0358
SOR	N/A	0.0997 ± 0.0235	N/A	0.1036 ± 0.0395
ROR	N/A	0.1177 ± 0.0240	N/A	0.1252 ± 0.0406
OURS-U	0.0754 ± 0.0351	0.0698 ± 0.0216	0.0800 ± 0.0423	0.0651 ± 0.0242
OURS	0.0565 ± 0.0244	0.0436 ± 0.0126	0.0610 ± 0.0335	0.0416 ± 0.0165

Note: Values are reported as mean ± SD.

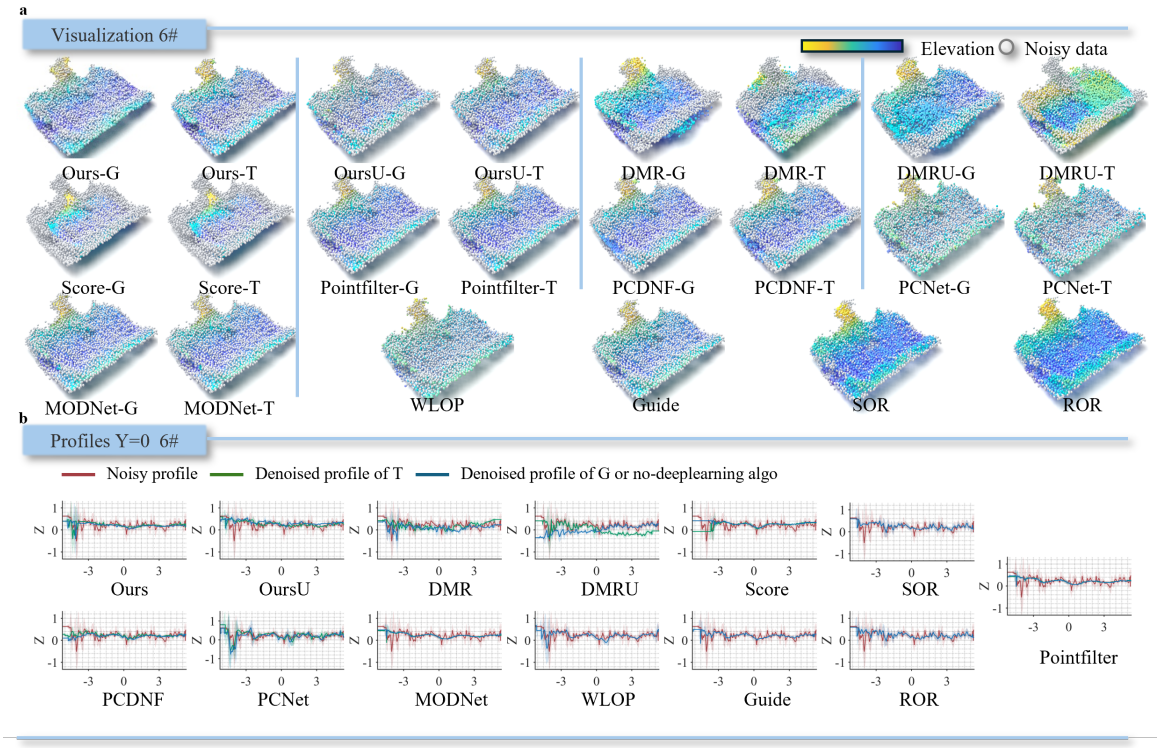

Figure 27: On-site performance and generalization comparison experiments (6#): a. Qualitative visualization on a representative sample; b. Comparison of the central Y-profile for the representative sample.

References

[1] L. Sun, Z. Huang, Y. Zhu, W. Xiong, C. Cai, Label-free defect detection of underwater piles from sonar point clouds via topology-aware pseudo supervision, Automation in Construction 190 (2026) 107133. doi:https://doi.org/10.1016/j.autcon.2026.107133.